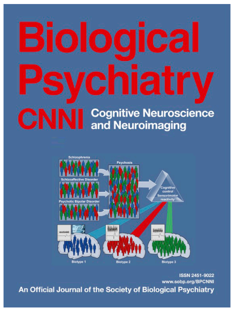
Accepted Manuscript



Interoception and Mental Health: a Roadmap

Sahib S. Khalsa, Ralph Adolphs, Oliver G. Cameron, Hugo D. Critchley, Paul W. Davenport, Justin S. Feinstein, Jamie D. Feusner, Sarah N. Garfinkel, Richard D. Lane, Wolf E. Mehling, Alicia E. Meuret, Charles B. Nemeroff, Stephen Oppenheimer, Frederike H. Petzschner, Olga Pollatos, Jamie L. Rhudy, Lawrence P. Schramm, W Kyle Simmons, Murray B. Stein, Klaas E. Stephan, Omer Van Den Bergh, Ilse Van Diest, Andreas von Leupoldt, Martin P. Paulus

PII: S2451-9022(17)30234-3

DOI: 10.1016/j.bpsc.2017.12.004

Reference: BPSC 225

To appear in: *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging*

Received Date: 23 August 2017

Revised Date: 20 November 2017

Accepted Date: 10 December 2017

Please cite this article as: Khalsa S.S., Adolphs R., Cameron O.G., Critchley H.D., Davenport P.W., Feinstein J.S., Feusner J.D., Garfinkel S.N., Lane R.D., Mehling W.E., Meuret A.E., Nemeroff C.B., Oppenheimer S., Petzschner F.H., Pollatos O., Rhudy J.L., Schramm L.P., Simmons W.K., Stein M.B., Stephan K.E., Van Den Bergh O., Van Diest I., von Leupoldt A., Paulus M.P. & the Interoception Summit 2016 participants, Interoception and Mental Health: a Roadmap, *Biological Psychiatry: Cognitive Neuroscience and Neuroimaging* (2018), doi: 10.1016/j.bpsc.2017.12.004.

Interoception and Mental Health: a Roadmap

Sahib S. Khalsa [1,2]*, Ralph Adolphs [3], Oliver G. Cameron [4], Hugo D. Critchley [5], Paul W. Davenport [6], Justin S. Feinstein [1,2], Jamie D. Feusner [7], Sarah N. Garfinkel [5], Richard D. Lane [8], Wolf E. Mehling [9], Alicia E. Meuret [10], Charles B. Nemeroff [11], Stephen Oppenheimer [12], Frederike H. Petzschner [13], Olga Pollatos [14], Jamie L. Rhudy [15], Lawrence P. Schramm [16,17], W Kyle Simmons [1,2], Murray B. Stein [18], Klaas E. Stephan [13], Omer Van Den Bergh [19], Ilse Van Diest [19], Andreas von Leupoldt [19], Martin P. Paulus [1], and the Interoception Summit 2016 participants [20].

Affiliations:

[1] Laureate Institute for Brain Research, Tulsa, Oklahoma, USA
[2] Oxley College of Health Sciences, University of Tulsa, Tulsa, Oklahoma, USA
[3] California Institute of Technology, Pasadena, California, USA
[4] Department of Psychiatry, University of Michigan, Ann Arbor, Michigan, USA
[5] Sackler Centre for Consciousness Science, University of Sussex, Brighton, UK
[6] Department of Physiology, University of Florida, Gainesville, Florida, USA
[7] Department of Psychiatry and Biobehavioral Sciences, Semel Institute for Neuroscience and Human Behavior, University of California at Los Angeles, Los Angeles, California, USA
[8] Department of Psychiatry, University of Arizona, Tucson, Arizona, USA
[9] Department of Family and Community Medicine, University of California San Francisco, San Francisco, California, USA
[10] Department of Psychology, Southern Methodist University, Dallas, Texas, USA
[11] Department of Psychiatry and Behavioral Sciences, University of Miami, Miami Florida, USA
[12] Department of Neurology, Johns Hopkins University, Baltimore, Maryland, USA
[13] Translational Neuromodeling Unit, Institute for Biomedical Engineering, University of Zurich, Zurich, Switzerland
[14] Department of Clinical and Health Psychology, Institute of Psychology and Education, Ulm University, Ulm, Germany
[15] Department of Psychology, University of Tulsa, Tulsa, Oklahoma, USA
[16] Department of Biomedical Engineering, Johns Hopkins University, Baltimore, Maryland, USA
[17] Department of Neuroscience, Johns Hopkins University, Baltimore, Maryland, USA
[18] Department of Psychiatry, University of California San Diego, San Diego, California, USA
[19] Department of Health Psychology, University of Leuven, Leuven, Belgium
[20] See acknowledgements and supplemental information

**Corresponding author*:
Sahib S. Khalsa, MD, PhD, Assistant Professor
Laureate Institute for Brain Research
Oxley College of Health Sciences
University of Tulsa
6655 S Yale Ave
Tulsa, OK 74136-3326
Phone: 918 502 5743
Fax: 918 502 5135
email: skhalsa@laureateinstitute.org

Keywords:
Interoception; mental health; biomarker; Research Domain Criteria (RDoC), computational psychiatry; treatment

Article information:
Running title: Interoception and mental health
Type of submission: Review
Number of words:
Abstract: 203 out of 250
Manuscript: 3994 out of 4000
Number of figures: 4
Number of tables: 3
Number of supplementary materials: 1
Number of references: 150 out of 150

Abstract

Interoception refers to the process by which the nervous system senses, interprets, and integrates signals originating from within the body, providing a moment-by-moment mapping of the body's internal landscape across conscious and unconscious levels. Interoceptive signaling has been considered a component process of reflexes, urges, feelings, drives, adaptive responses, cognitive and emotional experiences, highlighting its contributions to the maintenance of homeostatic functioning, body regulation, and survival. Dysfunction of interoception is increasingly recognized as an important component of different mental health conditions including anxiety disorders, mood disorders, eating disorders, addictive disorders, somatic symptom disorders, and others. However, a number of conceptual and methodological challenges have made it difficult for interoceptive constructs to be broadly applied in mental health research and treatment settings. In November 2016, the Laureate Institute for Brain Research organized the first Interoception Summit, a gathering of interoception experts from around the world, with the goal of accelerating progress in understanding the role of interoception in mental health. The discussions at the meeting were organized around four themes: interoceptive assessment, interoceptive integration, interoceptive psychopathology, and the generation of a roadmap that could serve as a guide for future endeavors. This article presents an overview of the emerging consensus generated by the meeting.

Introduction

Interoception refers collectively to the processing of internal bodily stimuli by the nervous system. Parcellation of the nervous system's processing of sensory signals into interoception, proprioception, and exteroception began more than a hundred years ago (1), although it was predated by interest in linking body-brain interactions with conscious experience (2, 3). Scientific interest in interoception has fluctuated (Figure 1A). In the 1980's biological psychiatry was inundated with observations of interoceptive disturbances in panic disorder (4-7), though the trend receded after it became clear that the etiological mechanism was broader than a single molecular receptor target (8). Recent years have witnessed a surge of interest on the topic of interoception, due in part to findings highlighting its integral role in emotional experience, self-regulation, decision-making, and consciousness. Importantly, interoception is not limited to conscious perception or even unique to the human species. From this perspective, interdisciplinary efforts to understand different features of interoception have been essential for advancing progress in cognitive and clinical neuroscience (Figure 1B).

< Insert Figure 1 about here >

I. Assessment

Body systems of interoception

Interoceptive processing occurs across all major biological systems involved in maintaining bodily homeostasis, including the cardiovascular (10, 11), pulmonary (12), gastrointestinal (13, 14), genitourinary (15), nociceptive (16), chemosensory (17), osmotic (18), thermoregulatory (19), visceral[1] (20), immune (24, 25), and autonomic systems (26, 27) (**Table 1**). There has been relatively little focus overall on the integration across bodily systems, thus it is not surprising that most investigations of the topic have been siloed within distinct research areas or scientific disciplines (see (28) and (29) for noteworthy exceptions).

< Insert Table 1 about here >

Features of interoception

Interoception is not a simple process but rather has several facets (30). The act of sensing, interpreting, and integrating information about the state of inner body systems can be related to different elements such as interoceptive attention, detection, discrimination, accuracy, insight, sensibility, and self-report (**Table 2**). However, most interoceptive processes occur outside the realm of conscious awareness. Consciously experienced elements are measured clinically via subjective report, and there are few observable interoceptive signs (e.g., heart rate, respiration rate, pupillary dilation, flushing, perspiration, piloerection, nociceptive reflexes) (**Table 3**). Experimental approaches can quantify different body systems and features of interoceptive processing. Nevertheless, these measures are only partially overlapping and likely reflect somewhat distinct neural

[1] Visceroception has classically referred to the perception of bodily signals arising specifically from visceral organs such as the heart, lungs, stomach, intestines, and bladder, along with other internal organs in the trunk of the body (20). It did not include organs like the skin or skeletal muscle, in contrast to contemporary definitions interoception which typically encompasses signals from both the viscera and all other tissues that relay a signal to the central nervous system about the current state of the body, including the skin and skeletal/smooth muscle fibers, via lamina I spinothalamic afferents (21-23).

processes (31). Access to the full range of interoceptive signals often involves invasive approaches, which tend to elicit physiological perturbations and index more objectively measurable features (32). However, many insights have been gained by the application of non-invasive approaches within neuroscience and psychological assessment contexts (33) (see "Eavesdropping on brain-body communication").

< Insert Table 2 about here >

Importance of an interoceptive taxonomy

There is no generally agreed-upon taxonomy for interoception science. Variable definitions have made it difficult to identify the features under investigation, let alone evaluate the quality of the findings. Based on the number of physiological systems involved, it could be questioned whether the terms "interoception" and "interoceptive awareness" are too broad. Interoceptive awareness is an umbrella term that was first used to describe a self-report subscale (34), but has subsequently been used to encompass any (or all) of the different interoception features accessible to conscious self-report. Researchers from different fields developed definitions that only partially overlapped reflecting the need for operationalization in neuroscience (35, 36) and clinical practice (9, 37). Here we develop a more coherent nomenclature for its various components (**Table 2**), mirroring developments in other fields, especially pain (38). One key aspect is the importance of distinguishing sensation (i.e., the "raw" signals conveyed by bodily sensors) from perception (39, 40). We return to this theme below.

< Insert Table 3 about here >

Multilevel investigations

While interoception research to date has typically focused on single organ systems, a expanded approach is called for that assesses multiple interoceptive organ systems or elements. Examples include targeting numerous interoceptive features simultaneously, or employing different tasks which converge upon the same feature (e.g., combining 'top-down' assessments of interoceptive attention with 'bottom-up' perturbation approaches in the same individual) (**Figure 2A**).

Sensing perturbations

The inner and outer worlds of the body constantly fluctuate. The nervous system monitors these environmental changes and responds adaptively in order to maintain a homeostatic balance and promote survival. As psychiatric disorders often promote or reflect the development of chronic homeostatic and allostatic disturbances (61) there is a need for methods capable of eliciting homeostatic perturbations in controlled settings, especially those assessing subjective and behavioral responses to valence and arousal deviations. However, interoception is not simply about afferent processing. The brain's constant monitoring of the body occurs in service of optimizing homeostatic regulation. This efferent limb is understudied (62), and paradigms which can effectively measure visceromotor outputs will be critical to establish sensitive assays of dysfunctional interoception and homeostatic regulation (e.g., detection of visceromotor-efferent neural signals controlling baroreflex sensitivity during modulation of visceral-afferent input by sympathetic drugs). The reliability and validity of methods should be rigorously established.

< Insert Figure 2 about here >

II. Integration

Interoception and domain specificity within the brain

There are fundamentally differing ways to interpret the evolution of brain and body signaling in humans. The processing of interoceptive input could be domain specific, with modular processing occurring in specialized, encapsulated neural circuits (e.g. cardiac, respiratory, urinary, genital, chemical, hormonal; see (63) for a review of domain specificity), or functionally coupled (e.g., cardio-respiratory, genito-urinary, chemo-hormonal) and integrated within a single neural circuit. Understanding the adaptive origins and functions of interoceptive domain specificity (if present) could tell us how the implementation and deployment of interoceptive signals by the nervous system contributes to disordered mental health. Since interoceptive signaling involves afferent and efferent inputs across multiple hierarchies within the authonomic and central nervous systems, identifying where and how information processing dysfunctions negatively impact mental health represents a challenging problem.

Neural pathways of interoception

Several pathways have been implicated in the neural processing of interoceptive signals, beginning with a rich interface between autonomic afferents and the central nervous system. Relay pathways involve primarily spinal, vagal, and glossopharyngeal afferents, with multiple levels of processing and integration in autonomic ganglia and spinal cord (11, 20, 23, 26). Several brainstem (nucleus of the solitary tract, parabrachial nucleus, periaqueductal gray), subcortical (thalamus, hypothalamus, hippocampus, amygdala), and cortical regions (insula and somatosensory cortices) represent key afferent processing regions (26, 64, 65). A complementary set of regions involved in visceromotor actions represent key efferent processing regions including the anterior insula, anterior cingulate, subgenual cingulate, orbitofrontal, ventromedial prefrontal, supplementary motor and premotor areas (66-68). It is noteworthy that these neural regions coincide closely with other sensory processing systems, especially the nociceptive and affective systems. The degree to which these represent distinct or overlapping systems is presently unclear.

Linking paradigms across units of analysis

A particular challenge when examining interoception is the fact that afferent sensory signals are integrated on several levels (peripherally, within the spinal cord, and supra-spinally) to form sets of interoceptive maps across different body systems. The brain appears to integrate information representing particular states of multiple systems simultaneously (cardiac, respiratory, chemical, hormonal, nociceptive etc.) (23), and it is imperative to be able to model and comparatively evaluate such mappings (**Figure 2B**). This poses many challenges. One approach might be to apply measures which assess multiple organ systems or interoceptive features simultaneously (see (64, 69, 70)), or to record activity across the brain, spinal cord and peripheral organs (71). However, it is also possible that multi-system assessments may reduce specificity for certain disorders, and may therefore be unnecessary. For example, some panic disorder patients may experience dyspnea but not palpitations. Localizing and then targeting the dysfunctional interoceptive domain would become more useful than broad multisystem interventions.

Timing and rhythm in interoceptive circuits

The physiological timescales and amplitudes of interoceptive signaling vary dramatically (e.g., heart rate (0.5-3.3 Hz), respiratory rate (0.08-1 Hz), gastric contractility (0.05-0.1 Hz), urinary frequency (0.000045-0.00012 Hz), with even slower changes in humoral mediators (Harrison & Savitz, special issue) (**Figure 2C and 2D**)). They also vary across individuals, and over the life-span (e.g. increased heart rates in infants/children). Despite the variance the brain tracks such changes in similar subregions including the insula, somatosensory cortices, cingulate, amygdala, thalamus and brainstem (64, 65, 72-74). Temporal synchrony or dys-synchrony between these systems may impact interoceptive experiences, affect, and behavior, although the exact mechanisms require further study (75). Repetitive events are another important element for learning, and while there are numerous classic studies on visceral learning at the peripheral organ system level (76, 77), we know little about the central mapping of learned visceral memories especially in psychiatric disorders (Simmons et al. special issue).

How can animal research improve the understanding of human interoceptive processing?

Although the inability to measure the subjective state of animals results in indirect inferences, well established tasks exist (e.g., conditioned interoceptive place preference (78) and odor aversion (79)). The principal utility of animal models is the hypothesis testing of mechanistic processes at the biological level independent of appraisal and cognition. These include examining effects of peripheral or central nervous system lesions on physiology/behavior, or mapping of peripheral/central interactions via stimulation of selective neurons/circuits using optogenetic methods (80, 81), and targeted gene expression manipulation to test genetic hypotheses (82). Animal models are advantageous in that they allow for identification of neural mechanisms that may be distinct from higher cognitive processes (e.g., non-mammalian (reptiles/birds) vs. mammalian (mice/rats/monkeys/apes/chimpanzees), invertebrate (octopus) vs. vertebrate (fish/monkeys). The study of interoception in non-human primates offers intriguing opportunities. Investigations in this area have been centered primarily on neural encoding of baroreceptor afferent stimulation (10) and neuroanatomical circuit tracing (83). Fewer studies have examined relationships between mechanistic manipulation of interoceptive experiences and neural representation in these animals (see (84, 85) for exceptions).

Eavesdropping on brain-body communications

Interoception is manifested by the conversation between the body and brain, via multiple afferent and efferent feedback loops (23, 86). Listening in on this process requires different approaches. Peripheral perturbations are often used to stimulate the afferent bottom-up transfer of information, usually of mechanical (32, 69, 73, 74), chemical (87-89), or hormonal origin (90) (**Supplementary Table 1**). Central perturbations to probe efferent top-down processes have most typically involved selective regulation of attentional focus (33, 91), and less commonly, expectancy manipulations such as placebo/sham delivery (46). Functional magnetic resonance imaging (92), positron emission tomography (93), and electroencephalography (94, 95), have provided the primary means of assessing neural circuitry. However, a host of novel tools are capable of inhibiting, stimulating, or modulating the activity of interoceptive brain networks. Non-invasive methods include the application of transcranial magnetic stimulation (96), transcranial direct and alternating current stimulation (97), low-intensity focused ultrasound (98), temporally interfering electric fields (99), transcutaneous vagus nerve stimulation (tVNS) (100), presentation of information during different phases of visceral rhythms (e.g., cardiac systole vs. diastole) (101), and assessment of cortiocardiac signaling (102). An important point is that many of the critical brain structures are difficult to modulate non-invasively as they are located deep within the brain or near the midline. Invasive measures do not share this limitation; and while their implementation is driven by clinical concerns, they can provide important insights. These include implanted VNS (103), direct brain stimulation (104), and intracranial electrode recordings (105, 106). Beyond these perturbation tools, the use of experimental methods to modulate expectancies, such as placebo and sham interventions, are key. These methods will help determine how sensitive psychiatric and other clinical patients' afferent/efferent feedback loops are to processes requiring integrations of environmental context with body-brain signals (illustrated in the next section). Finally, neurofeedback (e.g., fMRI/EEG) represents an exciting opportunity to participate in the brain-body conversation by simultaneously measuring and modulating brain regions during treatment (for a non-interoceptive example, see (107)). Equipped with these tools, the future looks promising, but to advance progress they need to be paired with better models of brain function.

Computational theories of interoception

Identifying the state of the body represents a problem that cannot be solved by pure sensing, since afferent signals from body sensors (interosensations) are not only noisy but often ambiguous (108). Recent computational theories suggest that interoception deploys Bayesian inference to address this challenge (39, 40, 66, 67, 109, 110) (**Figure 3 and 4**). Specifically, the brain is assumed to construct a so-called "generative model" of interosensations that combines a predictive mapping (from hidden bodily states to interosensations) with prior information (beliefs or expectations about bodily states, represented as probability distributions). This view is supported by findings that interoceptive perception is strongly shaped by expectations (23, 46, 111, 112), and by theoretical arguments that suggest Bayesian inference as a unifying principle for intero- and exteroception (40, 110).

< Insert Figure 3 about here >

Another argument supporting a Bayesian view on interoception is its relation to what constitutes arguably the brain's most fundamental task: the regulation (or control) of bodily states. Put simply, if the brain were unable to resolve the ambiguity of interosensations it would face difficulties in choosing appropriate actions to protect homeostasis. In information-theoretic terms, the challenge of keeping bodily states within narrow homeostatic ranges corresponds to choosing actions that minimize the long-term average Shannon surprise (entropy) of interosensations (39, 110). Solving this control problem requires knowledge or estimations of current and/or future bodily states and hence inference and predictions/forecasts—two natural domains of generative models.

Eliciting surprise-minimizing (homeostasis-restoring) actions changes the bodily state and thus interosensations. This means that inference and control of bodily states form a closed loop. Inference-control loops that minimize interoceptive surprise can be cast as hierarchical Bayesian models (HBMs). Anatomically, HBMs are plausible candidates given that interoceptive circuitry is structured hierarchically (67, 113). Under general assumptions, HBMs employ a small set of computational quantities – predictions, prediction errors, and precisions (40, 114). These quantities can support surprise-minimization in two ways: by adjusting beliefs (probability distributions) throughout the hierarchy (predictive coding (114)), or engaging actions that fulfil beliefs about bodily states (active inference (115)).

HBMs support both homeostatic (reactive) and allostatic (prospective) control. Reconsidering classical homeostatic setpoints as beliefs (i.e., probabilistic representations of expected/desired bodily states) enables reactive regulation at the bottom of the hierarchy (39, 110): here, prediction errors elicit reflex-like actions that minimize momentary interoceptive surprise. Allostatic regulation at longer time-scales is achieved through modulation of homeostatic beliefs by inferred or forecast states signaled from higher hierarchical levels (39). Importantly, belief precision determines the force/pace of corrective actions – that is, the tighter the expected range of bodily state, the more vigorous the elicited regulatory action. This offers a novel explanation for psychosomatic phenomena and placebo effects (40).

In summary, a hierarchical Bayesian perspective unifies interoception and homeostatic/allostatic control under the same computational principles. This provides a conceptual foundation for computational psychosomatics and supports a taxonomy of disease processes (40). One caveat is that the empirical evidence for hierarchical Bayesian principles of interoception and homeostatic/allostatic control is indirect so far. Studies designed to probe hierarchical Bayesian processes under experimentally controlled homeostatic perturbations will be crucial for finessing (or refuting) current computational concepts of interoception.

< Insert Figure 4 about here >

III. Psychopathology

Interoceptive psychopathology

Several conceptual and heuristic models have linked dysfunctions of interoception to mental health conditions. Specifically, mood and anxiety disorders have been linked to failures to appropriately anticipate changes in interoceptive states (116). Eating disorders show behavioral and neural abnormalities in interoceptive processing, particularly in the context of caloric anticipation (46, 47, 117, 118), although it remains unclear whether this is due to altered afferent signaling, altered central sensory processing, abnormal temperament, and/or metacognition. Drug addiction, another condition marked by interoceptive disturbances, has an overlapping neural circuitry and abnormal responses to interoceptive cues (50, 119-121). Interoceptive dysfunction also likely plays a role in conditions including posttraumatic stress disorder and somatic symptom disorders (9). Other disorders also have interoceptive symptom overlap, however, the specific feature involved may differ according to the disorder or affected individual (e.g., chronic pain (122, 123), Tourette's syndrome and other Tic disorders, borderline personality disorder, obsessive compulsive disorder, autism spectrum disorder (58), and functional developmental disorders (124)). **Table 3** lists diagnostic symptoms and clinical signs indicative of interoceptive dysfunction in several psychiatric disorders. Conditions which have a psychiatric component include fibromyalgia, chronic fatigue syndrome, irritable bowel syndrome (IBS), functional disorders within medicine (e.g., non-cardiac chest pain, functional dysphagia), as well as certain medical disorders (e.g., gastroesophageal reflux, asthma).

Alternatively, one can use a dimensional psychopathology approach to link processes underlying interoceptive dysfunction to psychiatric disorders. Transdiagnostic perspectives such as those provided by the research domain criteria (RDoC) (125) may be particularly helpful in identifying the potential role played by various interoceptive processes, as several of these may not be readily identified at the symptom report level relied upon by clinicians, and accordingly, may not have entered into the diagnostic specifications for DSM. This would allow for identification of mechanistic dysfunctions across units of analyses and might bridge the biological gap in current diagnostic classification frameworks by directly probing the links between physiological and psychological dysfunctions. Interoceptive investigations in mental health populations might reveal evidence of 1) attentional bias (e.g., hypervigilance), 2) distorted physiological sensitivity (e.g., blunted or heightened magnitude estimation in response to a perturbation), 3) cognitive bias (e.g., catastrophizing in response to an anticipated stimulus), 4) abnormal sensibility (e.g., tendency to label one's experiences in a particular way), and 5) impaired insight (e.g., poor confidence-accuracy correspondence on a task).

Determining whether interoceptive processes are a cause or consequence of developmental psychopathology, and which factors might impact this development (such as early life stress or pain), will be an important area for future research. Such studies may benefit from the examination of younger (126, 127) or older (128, 129) samples, premorbid identification and longitudinal tracking of individuals (130). Investigating the role of social cognition/theory of mind in clinically-relevant interoceptive inference generation represents another ripe opportunity (131).

Interoceptive tests and/or biomarkers

Since interoception is fundamentally a process linking body and brain, it is conceivable that objective measures of this process could serve as biological indicators of disease states. However, there is presently limited evidence for interoceptive predictors of diagnostic, prognostic, or treatment status (9, 132, 133). Biomarkers, such as those derived from neuroimaging or blood measurements, should be sensitive, specific, and unaffected by cognitive and emotional influences. However, it seems conceivable that the most clinically sensitive interoceptive measures might derive from probes which perturb physiological functions to engage specific metacognitive beliefs and/or expectations about bodily states. Such measures could facilitate differential diagnosis testing by revealing the presence of interoceptive dysfunction of biological (within a physiological system or systems), psychological (e.g., overly precise expectations about bodily states), or metacognitive (e.g., discrepant self-efficacy beliefs with regard to homeostatic/allostatic regulation) origin (40). This approach could be seen as analogous to a cardiac stress test, such that adequate engagement of the system under ecologically valid conditions is required in order to measure its dysfunction.

The most common application of interoceptive evaluation in current clinical practice occurs during interoceptive exposure psychotherapy for panic disorder (134). During this procedure patients self-induce varieties of interoceptive symptoms via low-arousal manipulations (hyperventilation, performing jumping jacks, spinning in a chair, breathing through a straw) while the clinician monitors their subjective distress level. Unfortunately these manipulations often fail to adequately reproduce the fear response, possibly because the patient retains full control over the stimulation (they can quit at any time) and the perturbation remains predictable with minimal uncertainty, raising the question of whether modulating both physiological homeostasis *and* the perception of controllability might further improve the ecological validity and efficacy of interoceptive exposures (135). A test to verify successful interoceptive exposure therapy for panic disorder involves completion of a standardized behavioral avoidance paradigm (136). In this setting, the degree of tolerance to being enclosed in a small dark chamber for ten minutes might provide behavioral evidence verifying tolerance to triggers of interoceptive dysregulation. There is also experimental evidence that pharmacological interoceptive exposure therapy can reduce anxiety disorder symptom severity either as monotherapy (7, 137-139) or as an augmentative approach (140). However, there are few studies of these procedures to date, the impact of such interventions on longer term outcomes (e.g., 6 months or beyond) are unknown, and none of these approaches have translated into clinical practice.

Current treatments relevant to interoception

Currently available therapies with an interoceptive basis include: 1) Pharmacotherapies directly modulating interoceptive physiology. Examples include adrenergic blockade (e.g., propranolol), or agonism (e.g., yohimbine), stimulants (e.g., methylphenidate), benzodiazepines, muscle relaxants, opioids, and others. 2) Cognitive behavioral therapy with exposure and response prevention to reverse or attenuate conditioned fears or form new learned associations. It is helpful in ameliorating cognitive biases in numerous disorders including depression, OCD, PTSD (specifically, prolonged exposure therapy), IBS, and chronic pain. Interoceptive exposure is a special example demonstrated to be highly effective in specific disorders (especially panic disorder). Behavioral activation therapy for depression sometimes includes exposure to experiences with positive interoceptive value. 3) Capnometry-assisted respiratory training. Based on the assumption that sustained hypocapnia resulting from hyperventilation is a key mechanism in the production and maintenance of panic, CO2 capnography assisted therapy aims to help patients voluntarily increase end-tidal partial pressure of carbon dioxide and tolerate physiological variability associated with panic attacks (141) (Meuret et al., special issue). 4) Mindfulness based stress reduction, yoga, and other meditation/movement based treatments may be aimed at improving metacognitive awareness of mind body connections, by systematically attending to sensations of breathing, cognitions, and/or other modulated body states (e.g., muscle stretching) (142).

Interoceptive treatments on the horizon

Several emerging technologies may have relevance for interoception and mental health.

1) Floatation-REST (Reduced Environmental Stimulation Therapy), an intervention which systematically attenuates exteroceptive sensory input to the nervous system, also appears to non-invasively enhance exposure to interoceptive sensations such as the breath and heartbeat (Feinstein et al., special issue). Preliminary data suggests that a single one-hour session has a short-term anxiolytic and antidepressant effect in patients with comorbid anxiety and depression (143), but further research is needed to evaluate the safety, feasibility, and potential for long-term efficacy in psychiatric populations.

2) Perturbation approaches. Minimally invasive tools capable of systematically modulating interoceptive processing, such as inspiratory breathing loads (144), core body thermomodulation (145, 146), and tVNS (147) are several approaches awaiting further investigation. Given the hypothesis of noisy baseline afferent signaling, these approaches may systematically enhance the signal-to-noise ratio and facilitate interoceptive learning. A key aspect in discerning clinical efficacy of any perturbation may be the extent to which the patient perceives controllability over the intervention, and is willing/able to

surrender this parameter in treatment. Interventions in which escape or active avoidance behaviors are directly measurable may provide especially meaningful information (Pane-Farre et al, special issue).

IV. Roadmap

The road ahead

Beyond the issues outlined previously, progress in determining the relevance of interoception for mental health relies upon emphasizing the features that distinguish it from other sensory modalities. Interoception seemingly involves a high degree of connectivity within the brain (148). It appears to be tightly linked to the self and survival through homeostatic maintenance of the body, and by helping us to represent how things are going in the present with respect to the experienced past and the anticipated future. These computations may depend on what has occurred to shape the body's internal landscape, and it is in this regard that learning, and malleability of representations over time, could play important roles.

The conceptual framework for investigating interoception may overlap with other processes, including emotion (149) and pain (150), as each is integral for maintaining bodily homeostasis. An important endeavor may involve the identification of which neural systems for interoception, emotion, cognition, and pain are overlapping, interdigitating, or even possibly identical. Additional effort is needed to define the neurophysiological nomenclature, core criteria, common features, developmental aspects, modulating factors, functional consequences, and putative pathophysiologic mechanisms of interoception in mental health disorders.

The present work offers some conceptual distinctions and some mutually agreed upon terminology, with many others still needed. Several “low hanging fruits” have been mentioned, as well as promising emerging technologies and tools. Further empirical work will be critical to delineate how interoception can be mapped to mental health measures, models, and approaches, and benchmarks for success/failure need to be established. Models of interoceptive processing have been described which improve upon the traditional stimulus, sensorimotor processing, and response function concepts, but these models remain theoretical and await further testing. The current document is therefore best viewed as a work in progress.

Acknowledgments: The authors would like to express their sincere appreciation to The William K. Warren Foundation for supporting the Interoception Summit 2016, and to all of the Laureate Institute for Brain Research (LIBR) staff for their assistance with facilitating the meeting. Interoception Summit 2016 participant co-authors: Vivien Ainley, Obada Al Zoubi, Robin Aupperle, Jason Avery, Leslie Baxter, Christoph Benke, Laura Berner, Jerzy Bodurka, Eric Breese, Tiffany Brown, Kaiping Burrows, Yoon-Hee Cha, Ashley Clausen, Danielle Deville, Laramie Duncan, Patrice Duquette, Hamed Ekhtiari, Thomas Fine, Bart Ford, Indira Garcia Cordero, Diamond Gleghorn, Yvette Guereca, Mahlega Hassanpour, Tanja Hechler, Aaron Heller, Natalie Hellman, Beate Herbert, Behnaz Jarrahi, Kara Kerr, Namik Kirlic, Megan Klabunde, Thomas Kraynak, Michael Kriegsman, Juliet Kroll, Rayus Kuplicki, Rachel Lapidus, Trang Le, Kyle Logie Hagen, Ahmad Mayeli, Amanda Morris, Nasir Naqvi, Kristina Oldroyd, Raquel Phillips, Tasha Poppa, Willliam Potter, Maria Puhl, Adam Safron, Margaret Sala, Jonathan Savitz, Heather Saxon, Will Schoenhals, Colin Stanwell-Smith, Adam Teed, Yuri Terasawa, Katie Thompson, Marisa Toups, Satoshi Umeda, Valerie Upshaw, Teresa Victor, Christina Wierenga, Colleen Wohlrab, Hung-wen Yeh, Adrian Yoris, Fadel Zeidan, Vadim Zotev, Nancy Zucker. SSK acknowledges support by The William K. Warren Foundation, a Brain and Behavior Foundation (formerly NARSAD) Young Investigator Award, the National Institute of Mental Health (NIMH) Award Number K23MH112949, and the National Institute of General Medical Sciences NIH/NIGMS P20GM121312. The content is solely the responsibility of the authors and does not necessarily represent the official views of the National Institutes of Health.

Disclosure statement: AEM reports the following disclosures: Research/Grants: National Institutes of Health (NIH) 1U01EB021952-01. Consulting (last three years): none. Stockholder: none. Scientific Advisory Boards: Anxiety and Depression Association of America. Board of Directors: none. Patents: none. Speakers Bureau: none. AvL reports the following disclosures: Research/Grants: Research Fund KU Leuven, Belgium (STRT/13/002 and DBOF/14/021); an infrastructure grant from the Herculesstichting, Belgium (AKUL/13/07); 'Asthenes' long-term structural funding Methusalem grant (METH/15/011) by the Flemish Government, Belgium. Consulting (last three years): none. Stockholder: none. Scientific Advisory Boards: none. Board of Directors: none. Patents: none. Speakers Bureau: none. CBN reports the following disclosures: Research/Grants: National Institutes of Health (NIH), Stanley Medical Research Institute. Consulting (last three years): Xhale, Takeda, Taisho Pharmaceutical Inc., Prismic Pharmaceuticals, Bracket (Clintara), Total Pain Solutions (TPS), Gerson Lehrman Group (GLG) Healthcare & Biomedical Council, Fortress Biotech, Sunovion Pharmaceuticals Inc., Sumitomo Dainippon Pharma, Janssen Research & Development LLC, Magstim, Inc., Navitor Pharmaceuticals, Inc., TC MSO, Inc. Stockholder: Xhale, Celgene, Seattle Genetics, Abbvie, OPKO Health, Inc., Bracket Intermediate Holding Corp., Network Life Sciences Inc., Antares. Scientific Advisory Boards: American Foundation for Suicide Prevention (AFSP), Brain and Behavior Research Foundation (BBRF) (formerly named National Alliance for Research on Schizophrenia and Depression [NARSAD]), Xhale, Anxiety and Depression Association of America (ADAA), Skyland Trail, Bracket (Clintara), RiverMend Health LLC, Laureate Institute

for Brain Research, Inc. Board of Directors: AFSP, Gratitude America, ADAA. Income sources or equity of $10,000 or more: American Psychiatric Publishing, Xhale, Bracket (Clintara), CME Outfitters, Takeda. Patents: Method and devices for transdermal delivery of lithium (US 6,375,990B1). Method of assessing antidepressant drug therapy via transport inhibition of monoamine neurotransmitters by ex vivo assay (US 7,148,027B2). Speakers Bureau: None. HDC reports the following disclosures: Research/Grants: European Research Council Horizon 2020 Proof of Concept Grant “HeartRater: Tools for the systematic evaluation of interoceptive ability.” Medical Research Council (UK) MRC Confidence in Concept Grant “Identifying neural, cognitive, and phenomenological markers of auditory verbal hallucinations in Borderline Personality.” MQ (Mental Health) PsyImpact “Aligning Dimensions of Interoceptive Experience (ADIE) to prevent development of anxiety disorders in autism.” Dr Mortimer and Theresa Sackler Foundation Sackler Centre for Consciousness Science, University of Sussex, BIAL foundation bursary “Microneurography as a tool for Consciousness Science.” Consulting (last three years): none. Stockholder: none. Scientific Advisory Boards: Emteq Ltd. Unpaid governor on board of charity “Reflecting Nature in art & science.” Board of Directors: Co-director of Sackler Centre for Consciousness Science, University of Sussex. KES reports the following disclosures: Research/Grants: German Research Council (DFG), Transregional Collaborative Research Centre, “Ingestive Behaviour: Homeostasis and Reward.” René and Susanne Braginsky Foundation. Consulting (last three years): none. Stockholder: none. Scientific Advisory Boards: none. Board of Directors: none. Patents: none. Speakers Bureau: none. IVD reports the following disclosures: Research/Grants: None. Consulting (last three years): none. Stockholder:

none. Scientific Advisory Boards: none. Board of Directors: none. Patents: none. Speakers Bureau: none. LPS reports the following disclosures: Scientific Advisory Board: Laureate Institute for Brain Research, Inc. JDF reports the following disclosures: Research/Grants: NIH R01MH105662, R21MH110865, and R01HD087712. Scientific Advisory Boards: International Obsessive-Compulsive Disorder Foundation Clinical and Scientific Advisory Board. JLR reports the following disclosures: Research/Grants: NIH R01MD007807 and the Oklahoma Center for the Advancement of Science and Technology (OCAST) HR15-079. JSF reports the following disclosures: Research/Grants: NIH/NIGMS P20GM121312, Brain and Behavior Foundation (formerly NARSAD) Young Investigator Award. Consulting (last three years): none. Stockholder: none. Scientific Advisory Boards: none. Board of Directors: none. Patents: none. Speakers Bureau: none. MBS reports the following disclosures: Scientific Advisory Board: Laureate Institute for Brain Research, Inc. Editorial Board: Depression and Anxiety, Biological Psychiatry. MPP reports the following disclosures: Research/Grants: The William K Warren Foundation and the NIH R01DA016663, NIH/NIGMS P20DA027834, NIH R01DA027797, NIH R01DA018307, NIH U01DA041089, and NIH 1R01MH101453. Consulting (last three years): has received royalties for an article about methamphetamine use disorder from UpToDate. OGC reports the following disclosures: Research/Grants: Research/Grants: none. Consulting (last three years): none. Stockholder: none. Scientific Advisory Boards: none. Board of Directors: none. Patents: none. Speakers Bureau: none. OVdB reports the following disclosures: Research/Grants: none. Consulting (last three years): none. Stockholder: none. Scientific Advisory Boards: Research Training Group 2271 of the Deutsche Forschungsgemeinschaft (DFG) on

"Expectation Maintenance vs. Change in the Context of Expectation Violations: Connecting Different Approaches", University of Marburg. Board of Directors: none. Patents: none. Speakers Bureau: none. RDL reports the following disclosures: Research/Grants: NIH. Consulting (last three years): none. Stockholder: none. Scientific Advisory Boards: none. Board of Directors: none. Patents: none. SNG reports the following disclosures: Research/Grants: none. Consulting (last three years): none. Stockholder: none. Scientific Advisory Boards: none. Board of Directors: none. Patents: none. Speakers Bureau: none. SSK reports the following disclosures: Research/Grants: NIH/NIMH K23MH112949, NIH/NIGMS P20GM121312, Brain and Behavior Foundation (formerly NARSAD) Young Investigator Award. Consulting (last three years): none. Stockholder: none. Scientific Advisory Boards: none. Board of Directors: none. Patents: none. Speakers Bureau: none. WEM reports the following disclosures: Research/Grants: NCIRE (Veterans Health Research Institute, San Francisco, California), Alzheimer’s Association, Mental Insight Foundation, The Pepper Foundation, UCSF Resource Allocation Program. WKS reports the following disclosures: Research/Grants: NIH P20GM121312, Brain and Behavior Foundation (formerly NARSAD) Young Investigator Award. Consulting (last three years): none. Stockholder: none. Scientific Advisory Boards: none. Board of Directors: none. Patents: none. Speakers Bureau: none. The remaining authors report no biomedical financial interests or potential conflicts of interest.

Non-painful
Cardiovascular, respiratory, gastrointestinal (esophageal, gastric, intestinal, colorectal), bladder, hunger, thirst, blood/serum (pH, osmolality, glucose), temperature, vasomotor flush, air hunger, muscle tension, shudder, itch, tickle, genital sensation, sensual touch, fatigue
Painful
Visceral: kidney stone, pleuritic, angina, pericardial, bowel ischemia, pelvic, sickle crisis Somatic: abscess/boil, bruising, myalgia, systemic/laceration inflammation, headache Skeletal: fractured/bruised bone, stress fracture, inflammatory/mechanical joint pain

Table 1. Physiological processes often ascribed to interoception. Several key distinctions are that interoceptive sensing 1) may be painful or non-painful, 2) occurs across the spectrums of high/low arousal and negative/positive valence, 3) usually occurs outside of conscious awareness (with the exception of pain sensations), and 4) is often (but not always) consciously experienced during instances of homeostatic perturbation.

Feature	Definition
Attention	Observing internal body sensations
Detection	Presence or absence of conscious report
Magnitude	Perceived intensity
Discrimination	Localize sensation to a specific channel or organ system, and differentiate it from other sensations
Accuracy (Sensitivity)	Correct and precise monitoring
Insight	Metacognitive evaluation of experience/performance (e.g., confidence-accuracy correspondence)
Sensibility	Self-perceived tendency to focus on interoceptive stimuli. Trait measure.
Self-report scales	Psychometric assessment via questionnaire. State/trait measure.

Table 2. Features of interoceptive awareness. For some examples of paradigms assessing each feature see **Supplementary Table 1**.

Psychiatric disorder	Symptom	Sign	Sample studies
Panic disorder	Palpitations, chest pain, dyspnea, choking, nausea, dizziness, flushing, derealization or depersonalization	Elevated heart rate, blood pressure, exaggerated escape, startle, and flinching	(41-43)
Depression	Increased or decreased appetite, fatigue, lethargy	Weight gain, weight loss, psychomotor slowing	(44, 45)
Eating disorders	Severe food restriction, hunger insensitivity, food anxiety, gastrointestinal complaints	Severe weight loss, binging, purging, compulsive exercise	(46, 47)
Somatic symptom disorders	Multiple current physical and nociceptive symptoms	Medical observations do not correspond with symptom report	(48, 49)
Substance use disorders	Physical symptoms associated with craving, intoxication, and/or withdrawal (drug specific)	Elevated/decreased: heart rate, respiratory rate, blood pressure, pupil dilation/ constriction, others (drug specific)	(50-52)
Posttraumatic stress disorder	Autonomic hypervigilance, derealization or depersonalization	Exaggerated startle, flinching, and escape responses, elevated heart rate, blood pressure	(53)
Generalized anxiety disorder	Muscle tension, headaches, fatigue, gastrointestinal complaints, pain	Trembling, twitching, shaking, sweating, nausea, exaggerated startle	(54, 55)
Depersonalization/ derealization disorder	Detachment from one's body, head fullness, tingling, light headedness	Physiological hyporeactivity to emotional stimuli	(56, 57)
Autism spectrum disorders	Skin hypersensitivity	Selective clothing preferences	(58-60)

Table 3. Diagnostic symptoms and clinical signs indicative of interoceptive dysfunction in some psychiatric disorders.

References:

1. Sherrington CS (1906): The integrative action of the nervous system. . New Haven, CT: Yale University Press.
2. Cameron OG (2002): Visceral sensory neuroscience: Interoception. *Visceral sensory neuroscience: Interoception*: Oxford University Press, New York, NY.
3. Adam G (1967): *Interoception and behaviour*. Akademiai Kiado.
4. Gorman JM, Fyer AF, Gliklich J, King D, Klein DF (1981): Effect of sodium lactate on patients with panic disorder and mitral valve prolapse. *The American Journal of Psychiatry*. 138:247-249.
5. Pohl R, Yeragani VK, Balon R, Rainey JM, Lycaki H, Ortiz A, et al. (1988): Isoproterenol-induced panic attacks. *Biological Psychiatry*. 24:891-902.
6. Woods SW, Charney DS, Loke J, Goodman WK, Redmond DE, Heninger GR (1986): Carbon dioxide sensitivity in panic anxiety. Ventilatory and anxiogenic response to carbon dioxide in healthy subjects and patients with panic anxiety before and after alprazolam treatment. *Archives of general psychiatry*. 43:900-909.
7. van den Hout MA, van der Molen GM, Griez E, Lousberg H, Nansen A (1987): Reduction of CO2-induced anxiety in patients with panic attacks after repeated CO2 exposure. *Am J Psychiatry*. 144:788-791.
8. Margraf J, Ehlers A, Roth WT (1986): Sodium lactate infusions and panic attacks: A review and critique. *Psychosomatic Medicine*. 48:23-51.
9. Khalsa SS, Lapidus RC (2016): Can Interoception Improve the Pragmatic Search for Biomarkers in Psychiatry? *Frontiers in psychiatry*. 7:121.
10. Oppenheimer S, Cechetto D (2016): The Insular Cortex and the Regulation of Cardiac Function. *Comprehensive Physiology*. 6:1081-1133.
11. Shivkumar K, Ajijola OA, Anand I, Armour JA, Chen PS, Esler M, et al. (2016): Clinical neurocardiology-defining the value of neuroscience-based cardiovascular therapeutics. *J Physiol*. 594:3911-3954.
12. von Leupoldt A, Chan PY, Esser RW, Davenport PW (2013): Emotions and neural processing of respiratory sensations investigated with respiratory-related evoked potentials. *Psychosom Med*. 75:244-252.
13. Mayer EA, Naliboff BD, Craig AD (2006): Neuroimaging of the brain-gut axis: from basic understanding to treatment of functional GI disorders. *Gastroenterology*. 131:1925-1942.
14. Kaye WH, Fudge JL, Paulus M (2009): New insights into symptoms and neurocircuit function of anorexia nervosa. *Nat Rev Neurosci*. 10:573-584.
15. Drake MJ, Fowler CJ, Griffiths D, Mayer E, Paton JF, Birder L (2010): Neural control of the lower urinary and gastrointestinal tracts: supraspinal CNS mechanisms. *Neurourology and urodynamics*. 29:119-127.

16. Simons LE, Elman I, Borsook D (2014): Psychological processing in chronic pain: a neural systems approach. *Neurosci Biobehav Rev*. 39:61-78.
17. Nattie E, Li A (2012): Central chemoreceptors: locations and functions. *Compr Physiol*. 2:221-254.
18. Stevenson RJ, Mahmut M, Rooney K (2015): Individual differences in the interoceptive states of hunger, fullness and thirst. *Appetite*. 95:44-57.
19. Fealey RD (2013): Interoception and autonomic nervous system reflexes thermoregulation. *Handbook of clinical neurology*. 117:79-88.
20. Janig W (1996): Neurobiology of visceral afferent neurons: neuroanatomy, functions, organ regulations and sensations. *Biol Psychol*. 42:29-51.
21. Cameron OG (2001): Interoception: The inside story - A model for psychosomatic processes. *Psychosomatic Medicine*. 63:697-710.
22. Olausson H, Lamarre Y, Backlund H, Morin C, Wallin BG, Starck G, et al. (2002): Unmyelinated tactile afferents signal touch and project to insular cortex. *Nat Neurosci*. 5:900-904.
23. Craig AD (2002): How do you feel? Interoception: the sense of the physiological condition of the body. *Nat Rev Neurosci*. 3:655-666.
24. Capuron L, Miller AH (2011): Immune system to brain signaling: neuropsychopharmacological implications. *Pharmacol Ther*. 130:226-238.
25. Irwin MR, Cole SW (2011): Reciprocal regulation of the neural and innate immune systems. *Nat Rev Immunol*. 11:625-632.
26. Critchley HD, Harrison NA (2013): Visceral influences on brain and behavior. *Neuron*. 77:624-638.
27. Dworkin BR (2000): Interoception. *Handbook of psychophysiology (2nd ed)*: Cambridge University Press, New York, NY, pp 1039-1506.
28. Tsakiris M, Critchley H (2016): Interoception beyond homeostasis: affect, cognition and mental health. *Philos Trans R Soc Lond B Biol Sci*. 371.
29. Lane RD, Wager TD (2009): Introduction to a special issue of Neuroimage on brain-body medicine. *Neuroimage*. 47:781-784.
30. Vaitl D (1996): Interoception. *Biological Psychology*. 42:1-27.
31. Baranauskas M, Grabauskaite A, Griskova-Bulanova I (2017): Brain responses and self-reported indices of interoception: Heartbeat evoked potentials are inversely associated with worrying about body sensations. *Physiol Behav*. 180:1-7.
32. Khalsa SS, Rudrauf D, Sandesara C, Olshansky B, Tranel D (2009): Bolus isoproterenol infusions provide a reliable method for assessing interoceptive awareness. *Int J Psychophysiol*. 72:34-45.
33. Critchley HD, Wiens S, Rotshtein P, Ohman A, Dolan RJ (2004): Neural systems supporting interoceptive awareness. *Nat Neurosci*. 7:189-195.

34. Garner DM, Olmstead MP, Polivy J (1983): Development and validation of a multidimensional eating disorder inventory for anorexia-nervosa and bulimiaA. *International Journal of Eating Disorders*. 2:15-34.

35. Garfinkel SN, Seth AK, Barrett AB, Suzuki K, Critchley HD (2015): Knowing your own heart: distinguishing interoceptive accuracy from interoceptive awareness. *Biol Psychol*. 104:65-74.

36. Forkmann T, Scherer A, Meessen J, Michal M, Schachinger H, Vogele C, et al. (2016): Making sense of what you sense: Disentangling interoceptive awareness, sensibility and accuracy. *Int J Psychophysiol*. 109:71-80.

37. Mehling W (2016): Differentiating attention styles and regulatory aspects of self-reported interoceptive sensibility. *Philos Trans R Soc Lond B Biol Sci*. 371.

38. IASP (2012): International Association for the Study of Pain Taxonomy. http://www.iasp-pain.org/Taxonomy?navItemNumber=576.

39. Stephan KE, Manjaly ZM, Mathys CD, Weber LA, Paliwal S, Gard T, et al. (2016): Allostatic Self-efficacy: A Metacognitive Theory of Dyshomeostasis-Induced Fatigue and Depression. *Front Hum Neurosci*. 10:550.

40. Petzschner FH, Weber LAE, Gard T, Stephan KE (2017): Computational Psychosomatics and Computational Psychiatry: Toward a Joint Framework for Differential Diagnosis. *Biol Psychiatry*.

41. Gorman JM, Kent J, Martinez J, Browne S, Coplan J, Papp LA (2001): Physiological changes during carbon dioxide inhalation in patients with panic disorder, major depression, and premenstrual dysphoric disorder: evidence for a central fear mechanism. *Archives of general psychiatry*. 58:125-131.

42. Pohl R, Yeragani VK, Balon R, Rainey JM, Lycaki H, Ortiz A, et al. (1988): Isoproterenol-induced panic attacks. *Biol Psychiatry*. 24:891-902.

43. Stein MB, Asmundson GJ (1994): Autonomic function in panic disorder: cardiorespiratory and plasma catecholamine responsivity to multiple challenges of the autonomic nervous system. *Biol Psychiatry*. 36:548-558.

44. Simmons WK, Burrows K, Avery JA, Kerr KL, Bodurka J, Savage CR, et al. (2016): Depression-Related Increases and Decreases in Appetite: Dissociable Patterns of Aberrant Activity in Reward and Interoceptive Neurocircuitry. *Am J Psychiatry*. 173:418-428.

45. Nierenberg AA, Pava JA, Clancy K, Rosenbaum JF, Fava M (1996): Are neurovegetative symptoms stable in relapsing or recurrent atypical depressive episodes? *Biol Psychiatry*. 40:691-696.

46. Khalsa SS, Craske MG, Li W, Vangala S, Strober M, Feusner JD (2015): Altered interoceptive awareness in anorexia nervosa: Effects of meal anticipation, consumption and bodily arousal. *The International journal of eating disorders*. 48:889-897.

47. Berner LA, Simmons AN, Wierenga CE, Bischoff-Grethe A, Paulus MP, Bailer UF, et al. (2017): Altered interoceptive activation before, during, and after aversive breathing load in women remitted from anorexia nervosa. *Psychol Med*.1-13.

48. Dimsdale JE, Creed F, Escobar J, Sharpe M, Wulsin L, Barsky A, et al. (2013): Somatic symptom disorder: an important change in DSM. *J Psychosom Res*. 75:223-228.

49. Barsky AJ, Peekna HM, Borus JF (2001): Somatic symptom reporting in women and men. *J Gen Intern Med*. 16:266-275.

50. Goldstein RZ, Craig AD, Bechara A, Garavan H, Childress AR, Paulus MP, et al. (2009): The neurocircuitry of impaired insight in drug addiction. *Trends Cogn Sci*. 13:372-380.

51. Naqvi NH, Bechara A (2010): The insula and drug addiction: an interoceptive view of pleasure, urges, and decision-making. *Brain structure & function*. 214:435-450.

52. Paulus MP, Stewart JL (2014): Interoception and drug addiction. *Neuropharmacology*. 76 Pt B:342-350.

53. Glenn DE, Acheson DT, Geyer MA, Nievergelt CM, Baker DG, Risbrough VB, et al. (2016): High and Low Threshold for Startle Reactivity Associated with Ptsd Symptoms but Not Ptsd Risk: Evidence from a Prospective Study of Active Duty Marines. *Depress Anxiety*. 33:192-202.

54. Pluess M, Conrad A, Wilhelm FH (2009): Muscle tension in generalized anxiety disorder: a critical review of the literature. *Journal of anxiety disorders*. 23:1-11.

55. Rapaport MH, Schettler P, Larson ER, Edwards SA, Dunlop BW, Rakofsky JJ, et al. (2016): Acute Swedish Massage Monotherapy Successfully Remediates Symptoms of Generalized Anxiety Disorder: A Proof-of-Concept, Randomized Controlled Study. *The Journal of clinical psychiatry*. 77:e883-891.

56. Sedeno L, Couto B, Melloni M, Canales-Johnson A, Yoris A, Baez S, et al. (2014): How Do You Feel when You Can't Feel Your Body? Interoception, Functional Connectivity and Emotional Processing in Depersonalization-Derealization Disorder. *PLoS One*. 9:e98769. doi: 98710.91371/journal.pone.0098769. eCollection 0092014.

57. Schulz A, Koster S, Beutel ME, Schachinger H, Vogele C, Rost S, et al. (2015): Altered patterns of heartbeat-evoked potentials in depersonalization/derealization disorder: neurophysiological evidence for impaired cortical representation of bodily signals. *Psychosom Med*. 77:506-516.

58. Garfinkel SN, Tiley C, O'Keeffe S, Harrison NA, Seth AK, Critchley HD (2016): Discrepancies between dimensions of interoception in autism: Implications for emotion and anxiety. *Biol Psychol*. 114:117-126.

59. Schauder KB, Mash LE, Bryant LK, Cascio CJ (2015): Interoceptive ability and body awareness in autism spectrum disorder. *J Exp Child Psychol*. 131:193-200.

60. Quattrocki E, Friston K (2014): Autism, oxytocin and interoception. *Neurosci Biobehav Rev*. 47:410-430.

61. Sterling P (2014): Homeostasis vs allostasis: implications for brain function and mental disorders. *JAMA Psychiatry*. 71:1192-1193.

62. Schulz A, Vogele C (2015): Interoception and stress. *Frontiers in psychology*. 6:993.

63. Spunt RP, Adolphs R (2017): A new look at domain specificity: insights from social neuroscience. *Nat Rev Neurosci*.

64. Hassanpour MS, Simmons WK, Feinstein JS, Luo Q, Lapidus R, Bodurka J, et al. (2017): The Insular Cortex Dynamically Maps Changes in Cardiorespiratory Interoception. *Neuropsychopharmacology*.

65. Khalsa SS, Rudrauf D, Feinstein JS, Tranel D (2009): The pathways of interoceptive awareness. *Nat Neurosci*. 12:1494-1496.

66. Seth AK, Suzuki K, Critchley HD (2011): An interoceptive predictive coding model of conscious presence. *Frontiers in psychology*. 2:395.

67. Barrett LF, Simmons WK (2015): Interoceptive predictions in the brain. *Nat Rev Neurosci*. 16:419-429.

68. Dum RP, Levinthal DJ, Strick PL (2016): Motor, cognitive, and affective areas of the cerebral cortex influence the adrenal medulla. *Proc Natl Acad Sci U S A*. 113:9922-9927.

69. von Leupoldt A, Sommer T, Kegat S, Baumann HJ, Klose H, Dahme B, et al. (2009): Dyspnea and pain share emotion-related brain network. *Neuroimage*. 48:200-206.

70. De Cort K, Schroijen M, Hurlemann R, Claassen S, Hoogenhout J, Van den Bergh O, et al. (2017): Modeling the development of panic disorder with interoceptive conditioning. *European neuropsychopharmacology : the journal of the European College of Neuropsychopharmacology*. 27:59-69.

71. Khan HS, Stroman PW (2015): Inter-individual differences in pain processing investigated by functional magnetic resonance imaging of the brainstem and spinal cord. *Neuroscience*. 307:231-241.

72. Gray MA, Taggart P, Sutton PM, Groves D, Holdright DR, Bradbury D, et al. (2007): A cortical potential reflecting cardiac function. *Proc Natl Acad Sci U S A*. 104:6818-6823.

73. Aziz Q, Thompson DG, Ng VW, Hamdy S, Sarkar S, Brammer MJ, et al. (2000): Cortical processing of human somatic and visceral sensation. *J Neurosci*. 20:2657-2663.

74. Wang GJ, Tomasi D, Backus W, Wang R, Telang F, Geliebter A, et al. (2008): Gastric distention activates satiety circuitry in the human brain. *Neuroimage*. 39:1824-1831.

75. Wittmann M (2009): The inner experience of time. *Philos Trans R Soc Lond B Biol Sci*. 364:1955-1967.

76. Ádám G (1998): Visceral perception: Understanding internal cognition. *The Plenum series in behavioral psychophysiology and medicine*: Plenum Press, New York, NY.

77. Dworkin BR (1993): *Learning and physiological regulation*. Chicago: The University of Chicago Press.

78. Schechter MD, Calcagnetti DJ (1998): Continued trends in the conditioned place preference literature from 1992 to 1996, inclusive, with a cross-indexed bibliography. *Neurosci Biobehav Rev*. 22:827-846.

79. Haroutunian V, Campbell BA (1979): Emergence of interoceptive and exteroceptive control of behavior in rats. *Science*. 205:927-929.

80. Williams EK, Chang RB, Strochlic DE, Umans BD, Lowell BB, Liberles SD (2016): Sensory Neurons that Detect Stretch and Nutrients in the Digestive System. *Cell*. 166:209-221.

81. Chang RB, Strochlic DE, Williams EK, Umans BD, Liberles SD (2015): Vagal Sensory Neuron Subtypes that Differentially Control Breathing. *Cell*. 161:622-633.

82. Halim D, Wilson MP, Oliver D, Brosens E, Verheij JB, Han Y, et al. (2017): Loss of LMOD1 impairs smooth muscle cytocontractility and causes megacystis microcolon intestinal hypoperistalsis syndrome in humans and mice. *Proc Natl Acad Sci U S A*. 114:E2739-E2747.

83. Mesulam MM, Mufson EJ (1982): Insula of the old world monkey. I. Architectonics in the insulo-orbito-temporal component of the paralimbic brain. *The Journal of comparative neurology*. 212:1-22.

84. Mitz AR, Chacko RV, Putnam PT, Rudebeck PH, Murray EA (2017): Using pupil size and heart rate to infer affective states during behavioral neurophysiology and neuropsychology experiments. *Journal of neuroscience methods*. 279:1-12.

85. Rudebeck PH, Putnam PT, Daniels TE, Yang T, Mitz AR, Rhodes SE, et al. (2014): A role for primate subgenual cingulate cortex in sustaining autonomic arousal. *Proc Natl Acad Sci U S A*. 111:5391-5396.

86. Damasio AR (1999): *The feeling of what happens: body and emotion in the making of consciousness*. New York: Harcourt Brace.

87. Hannestad J, Subramanyam K, Dellagioia N, Planeta-Wilson B, Weinzimmer D, Pittman B, et al. (2012): Glucose metabolism in the insula and cingulate is affected by systemic inflammation in humans. *Journal of nuclear medicine : official publication, Society of Nuclear Medicine*. 53:601-607.

88. Harrison NA, Brydon L, Walker C, Gray MA, Steptoe A, Critchley HD (2009): Inflammation causes mood changes through alterations in subgenual cingulate activity and mesolimbic connectivity. *Biol Psychiatry*. 66:407-414.

89. Harrison NA, Brydon L, Walker C, Gray MA, Steptoe A, Dolan RJ, et al. (2009): Neural origins of human sickness in interoceptive responses to inflammation. *Biological Psychiatry*. 66:415-422.

90. Schulz A, Strelzyk F, Ferreira de Sa DS, Naumann E, Vogele C, Schachinger H (2013): Cortisol rapidly affects amplitudes of heartbeat-evoked brain potentials--implications for the contribution of stress to an altered perception of physical sensations? *Psychoneuroendocrinology*. 38:2686-2693.

91. Avery JA, Drevets WC, Moseman SE, Bodurka J, Barcalow JC, Simmons WK (2014): Major depressive disorder is associated with abnormal interoceptive activity and functional connectivity in the insula. *Biological Psychiatry*. 76:258-266.

92. Schulz SM (2016): Neural correlates of heart-focused interoception: a functional magnetic resonance imaging meta-analysis. *Philos Trans R Soc Lond B Biol Sci*. 371.

93. Cameron OG, Huang GC, Nichols T, Koeppe RA, Minoshima S, Rose D, et al. (2007): Reduced gamma-aminobutyric acid(A)-benzodiazepine binding sites in insular cortex of individuals with panic disorder. *Archives of general psychiatry*. 64:793-800.

94. Pollatos O, Schandry R (2004): Accuracy of heartbeat perception is reflected in the amplitude of the heartbeat-evoked brain potential. *Psychophysiology*. 41:476-482.

95. von Leupoldt A, Keil A, Chan PY, Bradley MM, Lang PJ, Davenport PW (2010): Cortical sources of the respiratory-related evoked potential. *Respiratory physiology & neurobiology*. 170:198-201.

96. Pollatos O, Herbert BM, Mai S, Kammer T (2016): Changes in interoceptive processes following brain stimulation. *Philos Trans R Soc Lond B Biol Sci*. 371.

97. Fox KC, Christoff K (2015): Transcranial direct current stimulation to lateral prefrontal cortex could increase meta-awareness of mind wandering. *Proc Natl Acad Sci U S A*. 112:E2414.

98. Monti MM, Schnakers C, Korb AS, Bystritsky A, Vespa PM (2016): Non-Invasive Ultrasonic Thalamic Stimulation in Disorders of Consciousness after Severe Brain Injury: A First-in-Man Report. *Brain Stimul*. 9:940-941.

99. Grossman N, Bono D, Dedic N, Kodandaramaiah SB, Rudenko A, Suk HJ, et al. (2017): Noninvasive Deep Brain Stimulation via Temporally Interfering Electric Fields. *Cell*. 169:1029-1041 e1016.

100. Antonino D, Teixeira AL, Maia-Lopes PM, Souza MC, Sabino-Carvalho JL, Murray AR, et al. (2017): Non-invasive vagus nerve stimulation acutely improves spontaneous cardiac baroreflex sensitivity in healthy young men: A randomized placebo-controlled trial. *Brain Stimul*.

101. Azevedo RT, Garfinkel SN, Critchley HD, Tsakiris M (2017): Cardiac afferent activity modulates the expression of racial stereotypes. *Nature communications*. 8:13854.

102. Panitz C, Hermann C, Mueller EM (2015): Conditioned and extinguished fear modulate functional corticocardiac coupling in humans. *Psychophysiology*. 52:1351-1360.

103. Nemeroff CB, Mayberg HS, Krahl SE, McNamara J, Frazer A, Henry TR, et al. (2006): VNS therapy in treatment-resistant depression: clinical evidence and putative neurobiological mechanisms. *Neuropsychopharmacology*. 31:1345-1355.

104. Mazzola L, Mauguiere F, Isnard J (2017): Electrical Stimulations of the Human Insula: Their Contribution to the Ictal Semiology of Insular Seizures. *J Clin Neurophysiol*. 34:307-314.

105. Kern M, Aertsen A, Schulze-Bonhage A, Ball T (2013): Heart cycle-related effects on event-related potentials, spectral power changes, and connectivity patterns in the human ECoG. *Neuroimage*. 81:178-190.

106. Park HD, Bernasconi F, Salomon R, Tallon-Baudry C, Spinelli L, Seeck M, et al. (2017): Neural Sources and Underlying Mechanisms of Neural Responses to Heartbeats, and their Role in Bodily Self-consciousness: An Intracranial EEG Study. *Cereb Cortex*.1-14.

107. Young KD, Misaki M, Harmer CJ, Victor T, Zotev V, Phillips R, et al. (2017): Real-Time Functional Magnetic Resonance Imaging Amygdala Neurofeedback Changes Positive Information Processing in Major Depressive Disorder. *Biol Psychiatry*.

108. Petersen S, Schroijen M, Molders C, Zenker S, Van den Bergh O (2014): Categorical interoception: perceptual organization of sensations from inside. *Psychol Sci*. 25:1059-1066.

109. Seth AK, Critchley HD (2013): Extending predictive processing to the body: Emotion as interoceptive inference. *Behavioral and Brain Sciences*. 36:227-228.

110. Pezzulo G, Rigoli F, Friston K (2015): Active Inference, homeostatic regulation and adaptive behavioural control. *Prog Neurobiol*. 134:17-35.

111. Buchel C, Geuter S, Sprenger C, Eippert F (2014): Placebo analgesia: a predictive coding perspective. *Neuron*. 81:1223-1239.

112. Seth AK (2013): Interoceptive inference, emotion, and the embodied self. *Trends Cogn Sci*. 17:565-573.

113. Smith R, Thayer JF, Khalsa SS, Lane RD (2017): The hierarchical basis of neurovisceral integration. *Neurosci Biobehav Rev*. 75:274-296.

114. Friston K (2008): Hierarchical models in the brain. *PLoS Comput Biol*. 4:e1000211.

115. Friston K (2010): The free-energy principle: a unified brain theory? *Nat Rev Neurosci*. 11:127-138.

116. Paulus MP, Stein MB (2010): Interoception in anxiety and depression. *Brain structure & function*. 214:451-463.

117. Kerr KL, Moseman SE, Avery JA, Bodurka J, Zucker NL, Simmons WK (2016): Altered Insula Activity during Visceral Interoception in Weight-Restored Patients with Anorexia Nervosa. *Neuropsychopharmacology*. 41:521-528.

118. Frank GK (2015): Advances from neuroimaging studies in eating disorders. *CNS Spectr*. 20:391-400.

119. Naqvi NH, Bechara A (2009): The hidden island of addiction: the insula. *Trends in neurosciences*. 32:56-67.

120. Paulus MP, Stewart JL, Haase L (2013): Treatment approaches for interoceptive dysfunctions in drug addiction. *Frontiers in psychiatry*. 4:137.

121. Avery JA, Burrows K, Kerr KL, Bodurka J, Khalsa SS, Paulus MP, et al. (2017): How the Brain Wants What the Body Needs: The Neural Basis of Positive Alliesthesia. *Neuropsychopharmacology*. 42:822-830.

122. Hechler T, Endres D, Thorwart A (2016): Why Harmless Sensations Might Hurt in Individuals with Chronic Pain: About Heightened Prediction and Perception of Pain in the Mind. *Frontiers in psychology*. 7:1638.

123. Di Lernia D, Serino S, Riva G (2016): Pain in the body. Altered interoception in chronic pain conditions: A systematic review. *Neurosci Biobehav Rev*. 71:328-341.

124. Flack F, Pane-Farre CA, Zernikow B, Schaan L, Hechler T (2017): Do Interoceptive Sensations Provoke Fearful Responses in Adolescents With Chronic Headache or Chronic Abdominal Pain? A Preliminary Experimental Study. *J Pediatr Psychol*. 42:667-678.

125. Insel T, Cuthbert B, Garvey M, Heinssen R, Pine DS, Quinn K, et al. (2010): Research domain criteria (RDoC): toward a new classification framework for research on mental disorders. *Am J Psychiatry*. 167:748-751.

126. Koch A, Pollatos O (2014): Cardiac sensitivity in children: sex differences and its relationship to parameters of emotional processing. *Psychophysiology*. 51:932-941.

127. Maister L, Tang T, Tsakiris M (2017): Neurobehavioral evidence of interoceptive sensitivity in early infancy. *eLife*. 6.

128. Murphy J, Brewer R, Catmur C, Bird G (2017): Interoception and psychopathology: A developmental neuroscience perspective. *Dev Cogn Neurosci*. 23:45-56.

129. Khalsa SS, Rudrauf D, Tranel D (2009): Interoceptive awareness declines with age. *Psychophysiology*. 46:1130-1136.

130. von Leupoldt A, Mangelschots E, Niederstrasser NG, Braeken M, Billiet T, Van den Bergh BRH (2017): Prenatal stress exposure is associated with increased dyspnoea perception in adulthood. *Eur Respir J*. 50.

131. Ondobaka S, Kilner J, Friston K (2017): The role of interoceptive inference in theory of mind. *Brain Cogn*. 112:64-68.

132. Ehlers A (1995): A 1-year prospective study of panic attacks: clinical course and factors associated with maintenance. *J Abnorm Psychol*. 104:164-172.

133. Sundermann B, Bode J, Lueken U, Westphal D, Gerlach AL, Straube B, et al. (2017): Support Vector Machine Analysis of Functional Magnetic Resonance Imaging of Interoception Does Not Reliably Predict Individual Outcomes of Cognitive Behavioral Therapy in Panic Disorder with Agoraphobia. *Frontiers in psychiatry*. 8:99.

134. Craske MG, Barlow, D.H. (2007): *Mastery of your anxiety and panic: therapist guide*. 4th edition ed. Oxford: Oxford University Press.

135. Abelson JL, Khan S, Liberzon I, Erickson TM, Young EA (2008): Effects of perceived control and cognitive coping on endocrine stress responses to pharmacological activation. *Biol Psychiatry*. 64:701-707.

136. Richter J, Hamm AO, Pane-Farre CA, Gerlach AL, Gloster AT, Wittchen HU, et al. (2012): Dynamics of defensive reactivity in patients with panic disorder and agoraphobia: implications for the etiology of panic disorder. *Biol Psychiatry*. 72:512-520.

137. Beck JG, Shipherd JC, Zebb BJ (1997): How does interoceptive exposure for panic disorder work? An uncontrolled case study. *Journal of anxiety disorders*. 11:541-556.

138. Forsyth JP, Lejuez CW, Finlay C (2000): Anxiogenic effects of repeated administrations of 20% CO2-enriched air: stability within sessions and habituation across time. *J Behav Ther Exp Psychiatry*. 31:103-121.

139. Deacon B, Kemp JJ, Dixon LJ, Sy JT, Farrell NR, Zhang AR (2013): Maximizing the efficacy of interoceptive exposure by optimizing inhibitory learning: a randomized controlled trial. *Behav Res Ther*. 51:588-596.

140. Smits JA, Rosenfield D, Davis ML, Julian K, Handelsman PR, Otto MW, et al. (2014): Yohimbine enhancement of exposure therapy for social anxiety disorder: a randomized controlled trial. *Biol Psychiatry*. 75:840-846.

141. Meuret AE, Wilhelm FH, Ritz T, Roth WT (2008): Feedback of end-tidal pCO(2) as a therapeutic approach for panic disorder. *Journal of Psychiatric Research*. 42:560-568.

142. Farb N, Daubenmier J, Price CJ, Gard T, Kerr C, Dunn BD, et al. (2015): Interoception, contemplative practice, and health. *Frontiers in psychology*. 6:763.

143. Feinstein JS, Khalsa SS, Yeh H, Wohlrab C, Simmons WK, Stein MB, et al. (2017): Examining the short-term anxiolytic and antidepressant effect of Floatation-REST. *PLoS One*. In revision.

144. Van Diest I, Davenport P, Van den Bergh O, Miller S, Robertson E (2007): Interoceptive fearconditioning to an inspiratory load using 20% CO2 inhalation as an unconditioned stimulus. *Biological Psychology*. 75:211-211.

145. Janssen CW, Lowry CA, Mehl MR, Allen JJ, Kelly KL, Gartner DE, et al. (2016): Whole-Body Hyperthermia for the Treatment of Major Depressive Disorder: A Randomized Clinical Trial. *JAMA Psychiatry*. 73:789-795.

146. Kox M, van Eijk LT, Zwaag J, van den Wildenberg J, Sweep FC, van der Hoeven JG, et al. (2014): Voluntary activation of the sympathetic nervous system and attenuation of the innate immune response in humans. *Proc Natl Acad Sci U S A*. 111:7379-7384.

147. De Couck M, Cserjesi R, Caers R, Zijlstra WP, Widjaja D, Wolf N, et al. (2017): Effects of short and prolonged transcutaneous vagus nerve stimulation on heart rate variability in healthy subjects. *Autonomic neuroscience : basic & clinical*. 203:88-96.

148. Kleckner I.R. ZJ, Touroutoglou A., Chanes L., Xia C., Simmons W.K., Quigley K.S., Dickerson B.C., Barrett L.F. (2017): Evidence for a large-scale brain system supporting allostasis and interoception in humans *Nature Human Behavior*. 1:0069.

149. Anderson DJ, Adolphs R (2014): A framework for studying emotions across species. *Cell*. 157:187-200.

150. Kent ML, Tighe PJ, Belfer I, Brennan TJ, Bruehl S, Brummett CM, et al. (2017): The ACTTION-APS-AAPM Pain Taxonomy (AAAPT) Multidimensional Approach to Classifying Acute Pain Conditions. *The journal of pain : official journal of the American Pain Society*. 18:479-489.

Figure legends

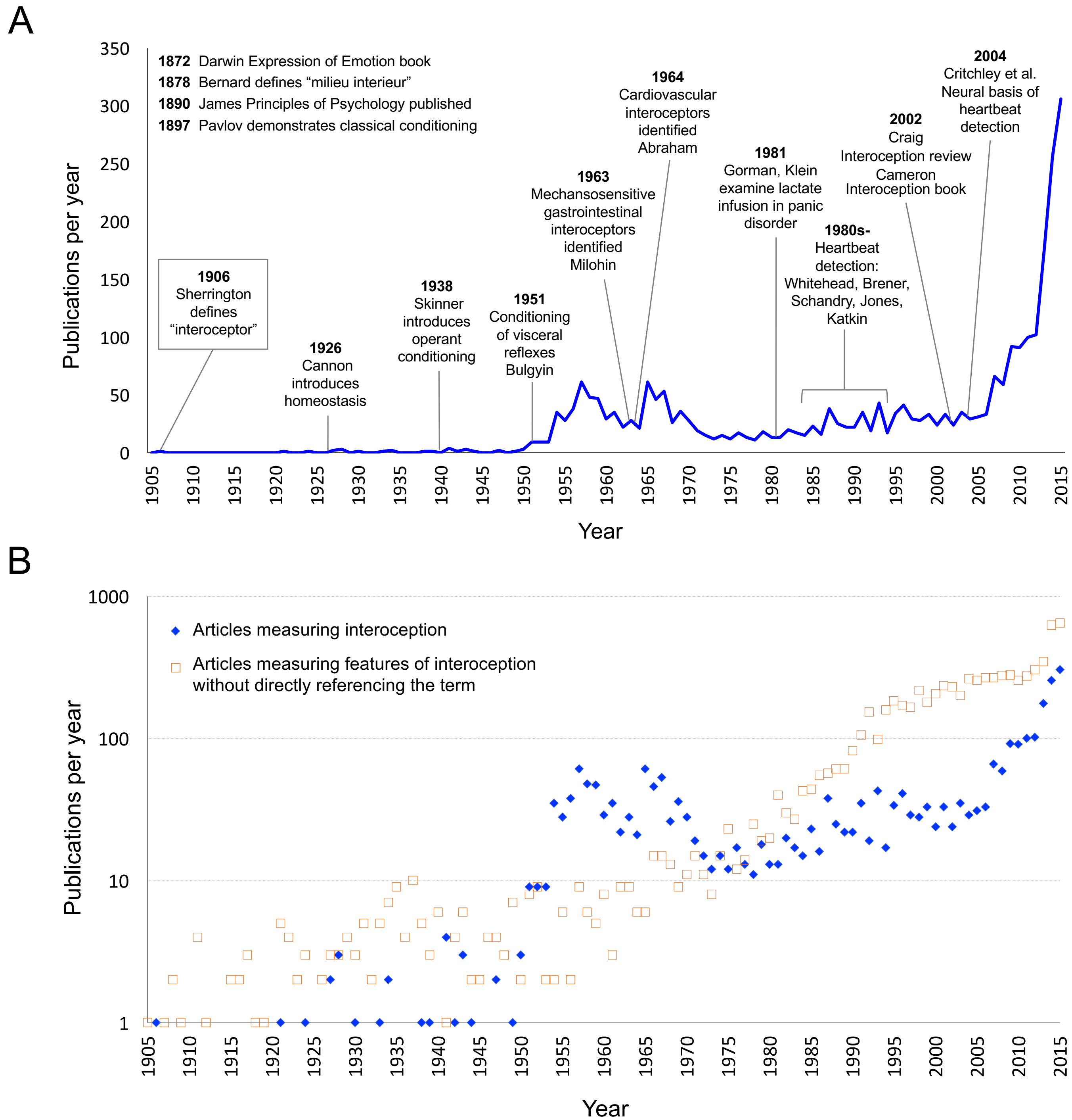
Figure 1. (A) Number of English language publications per year on interoception from Pubmed, Psychinfo and ISI Web of Knowledge. The timeline starts in 1905, one year before the publication of Charles Sherrington's book, "The integrative action of the nervous system," which first defined the concept of interoception. Key historical events relevant to interoception science are superimposed. (B) Publications per year on interoception, versus those investigating features of interoception that do not specifically refer to the term. These latter publications are more numerous, and arise mainly from basic neuroscience, physiology, and subspecialty disciplines within the biomedical field. Note the use of a logarithmic scale in the second panel. Figure reproduced and modified with permission from Khalsa & Lapidus (2016) (9).

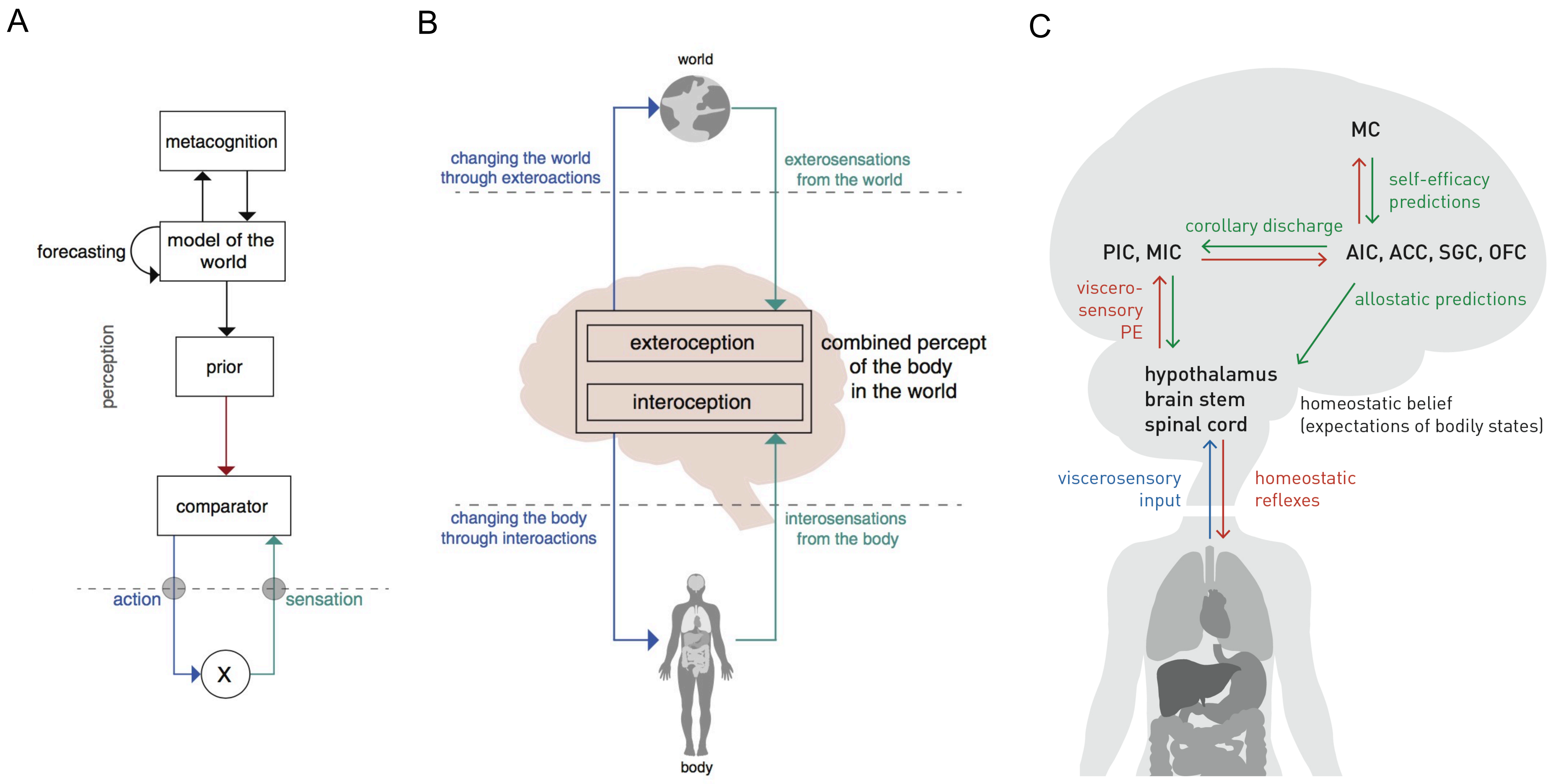
Figure 2. (A) Cardiac interoceptive processing measures and feature loading. This illustrates how the most commonly utilized heartbeat perception tasks differentially measure interoceptive features. Figure reproduced and modified with permission from Khalsa & Lapidus (2016) (9). (B) Multisystem interoceptive processing measures and feature loadings. Example of hypothetical approaches to measuring interoceptive processing across multiple systems. Approaches which perturb the state of the body are recommended, as well as convergent approaches, such as combined assessments of interoceptive attention and perturbation. (C) Central neural integration of interoceptive rhythms. Interoceptive rhythms vary considerably across the different systems of the

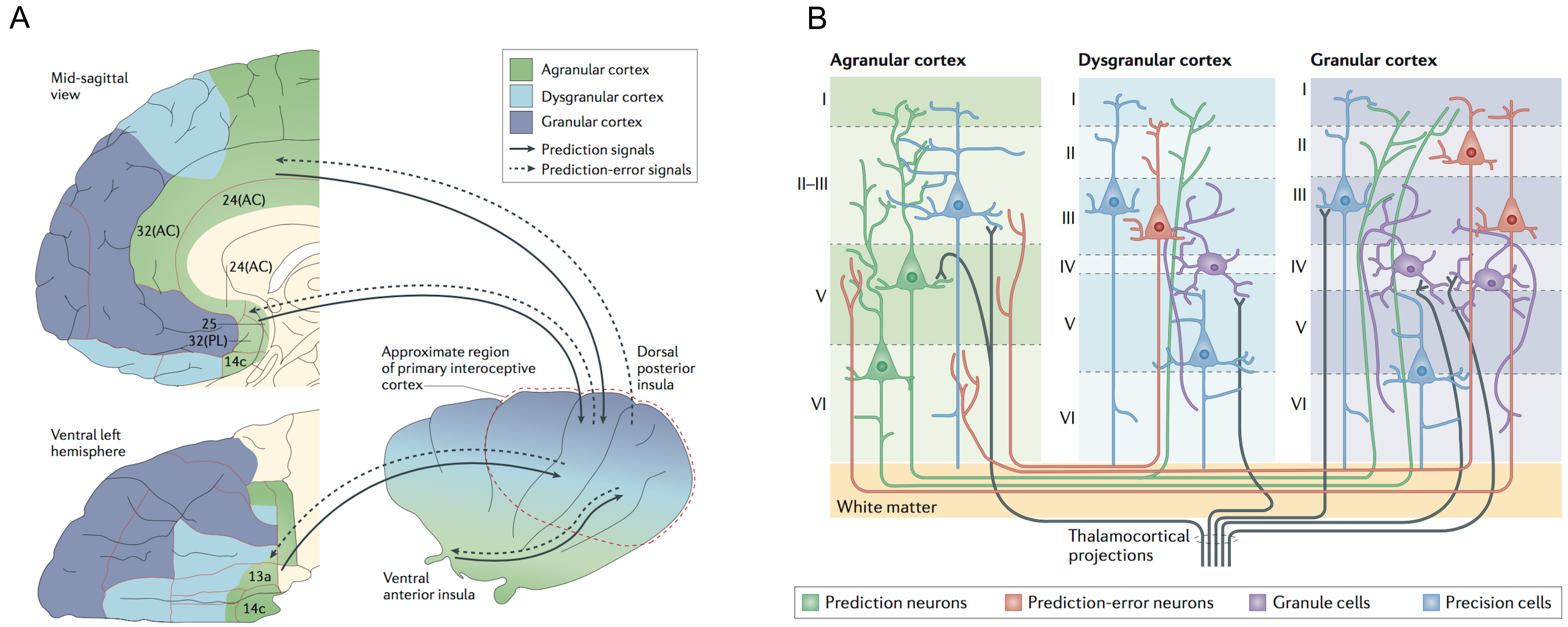
body. They exhibit complex characteristics, and are hierarchically integrated within discrete regions of the central nervous system. Figure modified with permission from Petzschner et al (2017) (39, 40). (D) Interoceptive rhythms vary in both amplitude and frequency. The top trace illustrates a hypothetical example of amplitude modulation signal superimposed on a static frequency. The middle trace illustrates a hypothetical frequency modulation signal superimposed on a static amplitude. The bottom trace illustrates a hypothetical signal change involving both amplitude and frequency modulations that are temporally correlated.

Figure 3. (A) Example of one possible form of a general inference-control loop illustrated within a hierarchical Bayesian model. (B) Highly schematic example of illustrating that both interoceptive and exteroceptive information are concurrently integrated to inform perceptual representations and action selection with respect to internally directed (e.g., visceromotor, autonomic) and externally directed (e.g., skeletomotor) actions. (C) General nodes which comprise a peripheral and central neural circuit for hierarchically integrating afferent interoceptive information into homeostatic reflexes, sensory and metacognitive representations, and allostatic regulators (predictions). ACC, anterior cingulate cortex; AIC, anterior insular cortex; MIC, mid-insular cortex; OFC, orbitofrontal cortex; PIC, posterior insular cortex; SGC, subgenual cortex. Figures reproduced, with permission, from Stephan et al (2016) (39) and Petzschner et al (2017) (40).

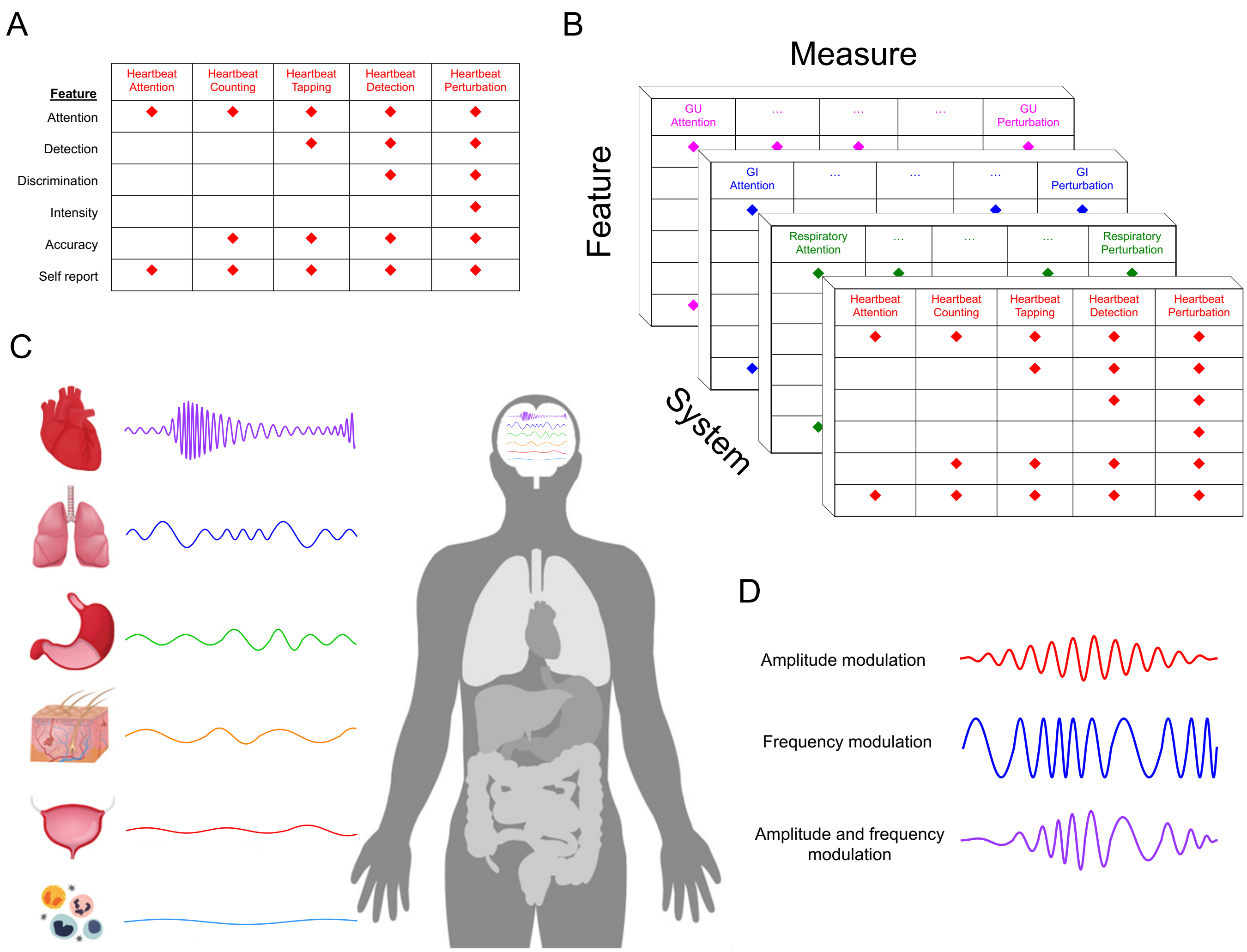
Figure 4. (A) Active inference implementation according to the Embodied Predictive Interoception Coding (EPIC) model. Agranular visceromotor cortices including the cingulate cortex, posterior ventral medial prefrontal cortex, posterior orbitofrontal cortex and the ventral anterior insula estimate the balance between autonomic, metabolic, and immunological resources available to the body, and its predicted requirements. These agranular visceromotor cortices issue allostatic predictions to hypothalamus, brainstem, and spinal cord nuclei to maintain a homeostatic internal milieu, and simultaneously to primary interoceptive sensory cortex in the mid and posterior insula. Interoceptive sensory cortex in the granular mid and posterior insula send reciprocal prediction error signals back to the agranular visceromotor regions to modify the predictions. Under usual circumstances these agranular regions are relatively insensitive to such feedback, which explains why interoceptive predictions are fairly stable in the face of body fluctuations. One hypothesis of the role of interoception in mental illness is that interoceptive input (i.e., posteriors) becomes increasingly decoupled from interoceptive predictions issued by agranular visceromotor cortex (priors), leading to increased interoceptive prediction error signals. This decoupling may present in the brain as “noisy afferent interoceptive inputs” (116). (B). Proposed intracortical architecture and intercortical connectivity for interoceptive predictive coding. Granular cortex contains six cell layers including granule cells, which are excitatory neurons that amplify and distribute thalamocortical inputs throughout the column. It is structurally similar to neocortex and therefore more recently evolved than the agranular and granular cortex. Within the insula, granular cortex is present in the mid and posterior sectors. AC, anterior cingulate; PL, prelimbic cortex. Figures reproduced, with permission, from Barrett & Simmons (2015) (67).


A
Publications per year
Year
1872 Darwin Expression of Emotion book
1878 Bernard defines "milieu interieur"
1890 James Principles of Psychology published
1897 Pavlov demonstrates classical conditioning
1906 Sherrington defines "interoceptor"
1926 Cannon introduces homeostasis
1938 Skinner introduces operant conditioning
1951 Conditioning of visceral reflexes Bulgyin
1963 Mechansosensitive gastrointestinal interoceptors identified Milohin
1964 Cardiovascular interoceptors identified Abraham
1981 Gorman, Klein examine lactate infusion in panic disorder
1980s- Heartbeat detection: Whitehead, Brener, Schandry, Jones, Katkin
2002 Craig Interoception review Cameron Interoception book
2004 Critchley et al. Neural basis of heartbeat detection
B
Articles measuring interoception
Articles measuring features of interoception without directly referencing the term
Publications per year
Year


A
metacognition
forecasting
model of the world
perception
prior
comparator
action
sensation
X
B
world
changing the world through exteroactions
exterosensations from the world
exteroception
interoception
combined percept of the body in the world
changing the body through interoactions
interosensations from the body
body
C
MC
self-efficacy predictions
corollary discharge
PIC, MIC
AIC, ACC, SGC, OFC
viscero-sensory PE
allostatic predictions
hypothalamus
brain stem
spinal cord
homeostatic belief
(expectations of bodily states)
viscerosensory input
homeostatic reflexes



A
Mid-sagittal view
Agranular cortex
Dysgranular cortex
Granular cortex
Prediction signals
Prediction-error signals
24(AC)
32(AC)
24(AC)
25
32(PL)
14c
Approximate region of primary interoceptive cortex
Dorsal posterior insula
Ventral left hemisphere
13a
14c
Ventral anterior insula
B
Agranular cortex
Dysgranular cortex
Granular cortex
I
II–III
V
VI
I
II
III
IV
V
VI
I
II
III
IV
V
VI
White matter
Thalamocortical projections
Prediction neurons
Prediction-error neurons
Granule cells
Precision cells


A
Feature
Attention
Detection
Discrimination
Intensity
Accuracy
Self report
Heartbeat Attention
Heartbeat Counting
Heartbeat Tapping
Heartbeat Detection
Heartbeat Perturbation
B
Measure
Feature
System
GU Attention
...
GU Perturbation
GI Attention
GI Perturbation
Respiratory Attention
Respiratory Perturbation
Heartbeat Attention
Heartbeat Counting
Heartbeat Tapping
Heartbeat Detection
Heartbeat Perturbation
C
D
Amplitude modulation
Frequency modulation
Amplitude and frequency modulation

Interoception and Mental Health: A Roadmap

Supplemental Information

Feature	Definition	Paradigms
Attention	Observing internal body sensations	C, GI (Simmons et al., 2013) R (Farb et al., 2013)
Detection	Presence or absence of conscious report	C (Khalsa et al., 2008) C (Garfinkel et al., 2013) R (Davenport et al., 2007) R (Paulus et al., 2012) GI (Holzl et al., 1996)
Magnitude	Perceived intensity	C, R (Khalsa et al., 2009b) R (Davenport et al., 2007) GI (Herbert et al, 2012b) GI (Naliboff et al., 2006) U (Jarrahi et al., 2015)
Discrimination	Localize sensation to a specific channel or organ system, and differentiate it from other sensations	C, R (Khalsa et al., 2015) GI (Aziz et al., 2000)
Accuracy (sensitivity)	Correct and precise monitoring	C (Schandry et al., 1993) C, R (Khalsa et al., 2009a) R (Daubenmier et al., 2013) GI (Zaman et al., 2016)
Insight	Metacognitive evaluation of experience/performance (e.g., confidence-accuracy correspondence)	(Khalsa et al., 2008) (Garfinkel et al., 2015)
Sensibility	Self-perceived tendency to focus on interoceptive stimuli. Trait measure.	(Garfinkel et al., 2015) (Garfinkel et al., 2016)
Self-report scales	Psychometric assessment via questionnaire. State/trait measure.	(Shields et al., 1989) (Porges, 1993) (Labus et al., 2007) (Mehling et al., 2012)

Supplementary Table S1. Features of interoceptive awareness. The illustrated paradigms span several physiological systems including the cardiac (C), respiratory (R), gastrointestinal (GI), and urinary (U) systems. Note that the paradigms and references listed here are not exhaustive; they are provided as illustrative examples. Many other approaches and paradigms exist, some of which are described in the main manuscript.

Interoception taxonomy

Interoception: The overall process of how the nervous system senses, integrates, stores, and represents information about the state of the inner body at conscious and unconscious levels. It is important to note that the contemporary definition of interoception is not synonymous with the term visceroception, but it does subsume it. Visceroception classically refers to the perception of bodily signals arising specifically from visceral organs such as the heart, lungs, stomach, intestines, and bladder, along with other internal organs in the trunk of the body (1). Visceroception does not include organs like the skin or skeletal muscle, whereas interoception encompasses both visceral signaling and more broadly relates to all physiological tissues that relay a signal to the central nervous system about the current state of the body, including the skin and skeletal/smooth muscle fibers, via lamina I spinothalamic afferents (2-4).

Interoceptive awareness: Conceptualizations of interoceptive awareness have been rather diverse. The term was originally introduced in 1983 by Garner and colleagues (5), via the development of the 'interoceptive awareness' subscale, part of a self-report measure intended to assess eating disorder symptom severity. The items on the subscale were derived based on the clinicians' experience, as well as prior empirical work on 'gastric perceptivity' in eating disorders (6). However, only 2 of the 10 items actually assessed interoception-related symptoms ("I get confused as to whether or not I am hungry" and "I feel bloated after eating a small meal"), with the remainder of the subscale preferentially measuring alexithymia (e.g., "I get confused about what emotion I am feeling"). Ensuing publications over the next 20 years referred to interoceptive awareness in this context (1983-2004, a total of 51 in PubMed). Following a report by Critchley et al. (2004) entitled "Neural systems supporting interoceptive awareness" (7), it has subsequently been used to encompass any (or all) of the different interoception features accessible to conscious self-report (2004-present, 288 publications in PubMed) (8). A more coherent and precise terminology has now been developed to describe its various components (Supplementary Table S1).

Interoceptive attention: The ability to direct attentional resources towards the source of internal body sensations. It can be captured (i.e., triggered involuntarily) in a stimulus dependent or 'bottom up' manner (9, 10), or shifted purposefully in a goal directed or 'top down' manner (11, 12). There is an important distinction about attentional 'styles', which may be viewed on a spectrum of anxiety driven, evaluative/avoidant on one extreme, to mindful, non-judgmental, and accepting on the other (13). The latter style tends to be cultivated during clinical applications of mindfulness and other forms of meditative practice.

Interoceptive detection: The ability to detect the presence or absence of a stimulus. It is a binary variable, similar to detection of a light source when it is switched on or off, or a judgment about whether a river is flowing or not.

Interoceptive magnitude: The intensity with which an internal bodily event is perceived. It is therefore a stimulus parameter reflecting the amount of signal. It is a continuous variable, e.g., reflecting how much dyspnea is present, and is typically gauged via subjective reports from the individual using rating scales, for example visual analogue scales and numerical rating scales. Magnitude estimation has been explained as a combination of prior expectations and current sensory input (14), emphasizing its relevance for Hierarchical Bayesian Models (HBMs) of interoception. It is difficult to measure changes in this feature in the absence of physiological perturbations (15).

Interoceptive discrimination: An individual's ability to localize sensation within a specific interoceptive system and to differentiate it from non-interoceptive sensations. It often requires the ability to locate sensations to specific regions within the body (as opposed to elsewhere inside the body), or relative to an external signal. Examples include the ability to distinguish a feeling of fullness after a meal from an irritating cough, or from the noise of a television in the background. It may also require the ability to separate different sensations from within the same interoceptive source, for example, as occurs when swallowing a food bolus (proximal vs. distal esophageal sensation and subsequent gastric deposition).

Interoceptive accuracy: The ability to precisely and correctly monitor changes in internal body state. It is synonymous with interoceptive sensitivity, and it has been the most commonly studied feature of interoception. Typical approaches involve the simultaneous measurement of an objective marker (e.g., heart rate, degree of inspiratory breathing load), the subjective experience of the individual (e.g., counted heart rate, detection or intensity of breathing difficulty), and subsequent calculation of the relationship between them. Subjective variables may be dichotomous (e.g., sensation present or absent, as in interoceptive detection) or continuous (e.g. how intense the stimulus is, as in interoceptive magnitude or discrimination). The interoceptive stimuli that are attended to may be acutely punctuated (as in the heartbeat) or continuous and prolonged (as in gut or bladder fullness). Interoceptive accuracy necessarily depends upon interoceptive attention given the reliance on attentional mechanisms for generating accuracy estimates. Common measurement examples include: 1) d' for heartbeat detection tasks (a signal detection metric (16, 17), cardiac error score for heartbeat counting tasks (18, 19), 2) cross correlations between heart rate and dial tracings of perceived intensity following adrenergic infusion (8, 20), 3) percent accuracy for detection of breathing occlusion (21), 4) intraclass correlations between bladder volumes and urinary urge (22), and 5) cross correlations between respiratory trace and slider tracings of the perceived phase and depth of respiration (23).

Interoceptive insight: The metacognitive measure detailing the correspondence between subjective experience and behavior. This is most typically evaluated by assessing the correspondence between accuracy and performance confidence on specific tasks. For example, in a study in which experienced meditators did not have different interoceptive accuracy on a heartbeat detection task, they exhibited differences in interoceptive self-report, reporting that the task was easier and their performance was improved relative to nonmeditators (24). In another study, performance on a heartbeat counting task was positively correlated with self-reported confidence estimates only in individuals with high but not low interoceptive accuracy scores (25) (note that replacement of the term 'interoceptive awareness' with 'interoceptive insight' in the prior study (25) is in keeping with the more coherent nomenclature suggested in this consensus statement).

Interoceptive sensibility: The self-perceived dispositional tendency to focus on interoceptive stimuli across daily life. Assessments of this trait measure inherently require an evaluation of self-perceived tendencies across broad time spans. This is commonly assessed by asking individuals to reflect on their autobiographical experience by answering questions such as 'To what extent do you believe you focus on and detect internal bodily sensations?' (25, 26). Mehling (13) has argued that the measurement of interoceptive sensibility is substantially more complex than the approach employed by Garfinkel et al. (25), particularly as the rating does not capture the regulatory and accepting/non-judgmental aspects of experience that are relevant for clinical settings. To address this complexity Mehling (13) developed the Multidimensional Assessment of Interoceptive Awareness (MAIA), which assesses 8-items via self-report (Noticing, Not-Distracting, Not-Worrying, Attention Regulation, Emotional Awareness, Self-Regulation, Body Listening, and Trusting) (see next section).

Interoceptive self-report scales: The ability to reflect upon one's autobiographical experiences of interoceptive states, make judgments about their outcomes, and describe them through verbal or motor responses. It is most commonly assessed experimentally using instruments or scales and there are a variety of them (25, 27-30). There are likely to be many constructs that can be identified within this level. For example, it may be possible to investigate dissociations between an individual's self-report of their interoceptive experiences versus their partner report (e.g., someone who knows the participant well and has had regular opportunities to interact with and observe them in a variety of situations, an approach that has been utilized for neurological samples (31)). Accordingly, the self-report component may be viewed as one of the most complex and nuanced features awaiting further investigation.

Supplementary References

1. Janig W (1996): Neurobiology of visceral afferent neurons: neuroanatomy, functions, organ regulations and sensations. *Biol Psychol*. 42:29-51.
2. Cameron OG (2001): Interoception: The inside story - A model for psychosomatic processes. *Psychosomatic Medicine*. 63:697-710.
3. Olausson H, Lamarre Y, Backlund H, Morin C, Wallin BG, Starck G, et al. (2002): Unmyelinated tactile afferents signal touch and project to insular cortex. *Nat Neurosci*. 5:900-904.
4. Craig AD (2002): How do you feel? Interoception: the sense of the physiological condition of the body. *Nat Rev Neurosci*. 3:655-666.
5. Garner DM, Olmstead MP, Polivy J (1983): Development and validation of a multidimensional eating disorder inventory for anorexia-nervosa and bulimiaA. *International Journal of Eating Disorders*. 2:15-34.
6. Coddington RD, Bruch H (1970): Gastric perceptivity in normal, obese and schizophrenic subjects. *Psychosomatics: Journal of Consultation and Liaison Psychiatry*. 11:571-579.
7. Critchley HD, Wiens S, Rotshtein P, Ohman A, Dolan RJ (2004): Neural systems supporting interoceptive awareness. *Nat Neurosci*. 7:189-195.
8. Khalsa SS, Craske MG, Li W, Vangala S, Strober M, Feusner JD (2015): Altered interoceptive awareness in anorexia nervosa: Effects of meal anticipation, consumption and bodily arousal. *The International journal of eating disorders*. 48:889-897.
9. Hassanpour MS, Simmons WK, Feinstein JS, Luo Q, Lapidus R, Bodurka J, et al. (2017): The Insular Cortex Dynamically Maps Changes in Cardiorespiratory Interoception. *Neuropsychopharmacology*.
10. von Leupoldt A, Sommer T, Kegat S, Baumann HJ, Klose H, Dahme B, et al. (2009): Dyspnea and pain share emotion-related brain network. *NeuroImage*. 48:200-206.
11. Simmons WK, Avery JA, Barcalow JC, Bodurka J, Drevets WC, Bellgowan P (2013): Keeping the body in mind: Insula functional organization and functional connectivity integrate interoceptive, exteroceptive, and emotional awareness. *Human Brain Mapping*. 34:2944-2958.
12. Farb NA, Segal ZV, Anderson AK (2013): Attentional modulation of primary interoceptive and exteroceptive cortices. *Cereb Cortex*. 23:114-126. doi: 110.1093/cercor/bhr1385. Epub 2012 Jan 1019.
13. Mehling W (2016): Differentiating attention styles and regulatory aspects of self-reported interoceptive sensibility. *Philos Trans R Soc Lond B Biol Sci*. 371.
14. Petzschner FH, Glasauer S, Stephan KE (2015): A Bayesian perspective on magnitude estimation. *Trends Cogn Sci*. 19:285-293.
15. Khalsa SS, Lapidus RC (2016): Can Interoception Improve the Pragmatic Search for Biomarkers in Psychiatry? *Frontiers in psychiatry*. 7:121.

16. Brener J, Liu X, Ring C (1993): A method of constant stimuli for examining heartbeat detection: comparison with the Brener-Kluvitse and Whitehead methods. *Psychophysiology*. 30:657-665.

17. Khalsa SS, Rudrauf D, Tranel D (2009): Interoceptive awareness declines with age. *Psychophysiology*. 46:1130-1136.

18. Herbert BM, Ulbrich P, Schandry R (2007): Interoceptive sensitivity and physical effort: implications for the self-control of physical load in everyday life. *Psychophysiology*. 44:194-202.

19. Koch A, Pollatos O (2014): Cardiac sensitivity in children: sex differences and its relationship to parameters of emotional processing. *Psychophysiology*. 51:932-941.

20. Khalsa SS, Rudrauf D, Feinstein JS, Tranel D (2009): The pathways of interoceptive awareness. *Nat Neurosci*. 12:1494-1496.

21. Zhao W, Martin AD, Davenport PW (2003): Magnitude estimation of inspiratory resistive loads by double-lung transplant recipients. *1985)*. 94:576-582. Epub 2002 Oct 2011.

22. Heeringa R, van Koeveringe GA, Winkens B, van Kerrebroeck PE, de Wachter SG (2011): Degree of urge, perception of bladder fullness and bladder volume--how are they related? *J Urol*. 186:1352-1357. doi: 1310.1016/j.juro.2011.1305.1050.

23. Daubenmier J, Sze J, Kerr CE, Kemeny ME, Mehling W (2013): Follow your breath: respiratory interoceptive accuracy in experienced meditators. *Psychophysiology*. 50:777-789.

24. Khalsa SS, Rudrauf D, Damasio AR, Davidson RJ, Lutz A, Tranel D (2008): Interoceptive awareness in experienced meditators. *Psychophysiology*. 45:671-677.

25. Garfinkel SN, Seth AK, Barrett AB, Suzuki K, Critchley HD (2015): Knowing your own heart: distinguishing interoceptive accuracy from interoceptive awareness. *Biol Psychol*. 104:65-74.

26. Garfinkel SN, Tiley C, O'Keeffe S, Harrison NA, Seth AK, Critchley HD (2016): Discrepancies between dimensions of interoception in autism: Implications for emotion and anxiety. *Biol Psychol*. 114:117-126.

27. Labus JS, Mayer EA, Chang L, Bolus R, Naliboff BD (2007): The central role of gastrointestinal-specific anxiety in irritable bowel syndrome: further validation of the visceral sensitivity index. *Psychosom Med*. 69:89-98.

28. Shields S, M.E. M, A. S (1989): The body awareness questionnaire: reliability and validity. *J Pers Assess*. 53:802-815.

29. Porges S (1993): *Body perception questionnaire*. University of Maryland.

30. Mehling WE, Price C, Daubenmier JJ, Acree M, Bartmess E, Stewart A (2012): The Multidimensional Assessment of Interoceptive Awareness (MAIA). *PLoS One*. 7:e48230.

31. Barrash J, Asp E, Markon K, Manzel K, Anderson SW, Tranel D (2011): Dimensions of personality disturbance after focal brain damage: investigation with the Iowa Scales of Personality Change. *J Clin Exp Neuropsychol*. 33:833-852.

Interoception summit 2016 participant author appendix:

First Name	**Last Name**	**Institution 1**	**Institution 2**
Vivien	Ainley	Royal Holloway University	
Obada	Al Zoubi	University of Oklahoma	Laureate Institute for Brain Research
Robin	Aupperle	Laureate Institute for Brain Research	
Jason	Avery	Laureate Institute for Brain Research	
Leslie	Baxter	Barrow Neurological Institute	
Christoph	Benke	University of Greifswald	
Laura	Berner	University of California San Diego	
Jerzy	Bodurka	Laureate Institute for Brain Research	
Eric	Breese	Laureate Institute for Brain Research	
Tiffany	Brown	Unversity of California, San Diego	
Kaiping	Burrows	Laureate Institute for Brain Research	
Yoon-Hee	Cha	Laureate Institute for Brain Research	
Ashley	Clausen	Laureate Institute for Brain Research	
Kelly	Cosgrove	Laureate Institute for Brain Research	
Danielle	Deville	Laureate Institute for Brain Research	
Laramie	Duncan	Stanford University	
Patrice	Duquette	Patrice Duquette MDPC	
Hamed	Ekhtiari	Laureate Institute for Brain Research	
Thomas	Fine	University of Toledo	
Bart	Ford	Laureate Institute for Brain Research	
Indira	Garcia Cordero	Buenos Aires University	
Diamond	Gleghorn	Laureate Institute for Brain Research	
Yvette	Guereca	The University of Tulsa	
Mahlega	Hassanpour	Laureate Institute for Brain Research	
Tanja	Hechler	University of Trier	
Aaron	Heller	University of Miami	
Natalie	Hellman	The University of Tulsa	
Beate	Herbert	University of Teubingen	
Behnaz	Jarrahi	Stanford University School of Medicine	
Kara	Kerr	University of Tulsa	
Namik	Kirlic	Laureate Institute for Brain Research	
Megan	Klabunde	Stanford University	
Thomas	Kraynak	University of Pittsburgh	
Michael	Kriegsman	The University of Texas at Dallas	
Juliet	Kroll	Southern Methodist University	
Rayus	Kuplicki	Laureate Institute for Brain Research	

Rachel	Lapidus	Laureate Institute for Brain Research	
Trang	Le	University of Tulsa	
Kyle	Logie Hagen	University of Toronto	
Ahmad	Mayeli	University of Oklahoma	
Amanda	Morris	Oklahoma State University	
Nasir	Naqvi	Columbia University	
Kristina	Oldroyd	University of Utah	
Raquel	Phillips	Laureate Institute for Brain Research	
Tasha	Poppa	University of Southern California	
William	Potter	University of Tulsa	
Maria	Puhl	Laureate Institute for Brain Research	
Adam	Safron	Northwestern University	
Margaret	Sala	Southern Methodist University	
Jonathan	Savitz	Laureate Institute for Brain Research	
Heather	Saxon	University of Missouri	Dalton Cardiovascular Research Center
Will	Schoenhals	University of Tulsa	Laureate Institute for Brain Research
Colin	Stanwell-Smith	Floataway Inc.	
Adam	Teed	University of Texas at Dallas	
Yuri	Terasawa	Keio University	
Katie	Thompson	Laureate Institute for Brain Research	
Marisa	Toups	University of Texas Southwestern	
Satoshi	Umeda	Keio University	
Valerie	Upshaw	Laureate Institute for Brain Research	
Teresa	Victor	Laureate Institute for Brain Research	
Christina	Wierenga	University of California San Diego	
Colleen	Wohlrab	Laureate Institute for Brain Research	
Hung-wen	Yeh	Laureate Institute for Brain Research	
Adrian	Yoris	Favaloro University	
Fadel	Zeidan	Wake Forrest University	
Vadim	Zotev	Laureate Institute for Brain Research	
Nancy	Zucker	Duke University	